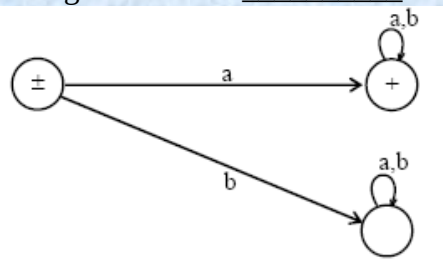
FINALTERM EXAMINATION
Fall 2012
CS402- Theory of Automata

Question No: 1 (Marks: 1) - Please choose one
If Σ = {aa, bb} , then Σ* will not contain
►**aaabbb**
►aabbbb
►aabbaa
►bbaabbbb

Question No: 2 (Marks: 1) - Please choose one
Below given FA has _____ RE.



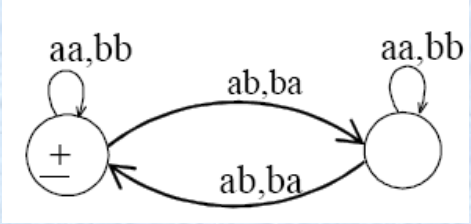
►**a(a+b)*** **(Page 14)**
► (a(a+b)*)*
►a(a+b)*a
►a(a+b)*a + b(a+b)*b

Question No: 3 (Marks: 1) - Please choose one
"One language can have _____ TG's".
►Only one
►Only two
►**More than one**
►Only three

دنیا میں سب سے مشکل کام اپنی اصلاح اور سب سے آسان کام دوسروں پر نکتہ چینی کرنا ہے

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Question No: 4 (Marks: 1) - Please choose one



Above given TG represents the language i.e.

►**EVEN-EVEN (Page 22)**
►PALINDROME
►FACTORIAL
►None of these

Question No: 5 (Marks: 1) - Please choose one

According to 1st part of the Kleene's theorem, If a language can be accepted by an FA then it can be accepted by a _____ as well.

►FA
►CFG
►GTG
►**TG (Page 25)**

Question No: 6 (Marks: 1) - Please choose one

Even-palindrome is a _____ language.

►**Non-regular** click here for detail
►Regular
►Regular but infinite
►Regular but finite

Question No: 7 (Marks: 1) - Please choose one

If L is a regular language then, L^c is also a _____ language.

► **Regular (Page 66)**
► Non-regular
► Regular but finite
► None of the given

Question No: 8 (Marks: 1) - Please choose one

Pumping lemma is generally used to prove that:

► A given language is infinite
► **A given language is not regular** Click here for detail
► Whether two given regular expressions of a regular language are equivalent or not
► None of these

بری صحبت سے تنہائی بہتر ہے اور تنہائی سے نیک صحبت بہتر ہے

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Question No: 9 (Marks: 1) - Please choose one
If the FA has N states, then test the words of length less than N. If no word is accepted by this FA, then it will _____ word/words.

►accept all
►**accept no (Page 85)**
►accept some
►reject no

Question No: 10 (Marks: 1) - Please choose one
In CFG, the symbols that can't be replaced by anything are called___.

► **Terminal (Page 87)**
► Non-Terminal
► Production
► All of given

Question No: 11 (Marks: 1) - Please choose one
Which of the following is a regular language?

► **String of odd number of zeroes** Click here for detail
► Set of all palindromes made up of 0's and 1's
► String of 0's whose length is a prime number
► All of these

Question No: 12 (Marks: 1) - Please choose one
Which of the following pairs of regular expressions are equivalent?

►1(001)* and (10)*10
►**x(xx)* and (x)*x**
► X^+ and X*
► X^+ and X* X^+

Question No: 13 (Marks: 1) - Please choose one
An alphabet of Σ is valid if

►No letter of Σ appears in middle of any other letter
►No letter of Σ appears at end of any other letter
►**No letter of Σ appears at start of any other letter (Page 4)**
►No letter of Σ appears at end or middle of any other letter



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Question No: 14 (Marks: 1) - Please choose one
Which of the following statement is true

►The length of the output string is greater than length of input string in moore machine. Click here for detail
►The length of the output string is greater than length of input string in mealy machine.
►The length of the output string is equal to length of input string in moore machine.
►The length of the output string is less than length of input string in mealy machine.

Question No: 15 (Marks: 1) - Please choose one
If a CFG has only productions of the form
nonterminal → string of two nonterminals
or
nonterminal → one terminal
then the CFG is said to be in _____

►**Chomsky Normal Form (Page 101)**
►Ambiguous Form
►Left Aligned Form
►Right Aligned Form

Question No: 16 (Marks: 1) - Please choose one
We can also represent an FA using different states e.g Accept state; Reject state, Read state etc.
The _____ state behaves as final state of an FA
►**Accept (Page 105)**
►Pop
►Push
►Reject

Question No: 17 (Marks: 1) - Please choose one
where the input string is placed before it is run, is called _____
►Date tape
►**Input Tape (Page 105)**
►Output Tape
►Magnetic tape

Question No: 18 (Marks: 1) - Please choose one
An FSM can be considered as TM

►Of finite tape length, rewinding capability and unidirectional tape movement
►Of finite tape length, without rewinding capability and bidirectional tape movement
►Of finite tape length, rewinding capability and bidirectional tape movement
►**Of finite tape length, without rewinding capability and unidirectional tape movement** click here for detail

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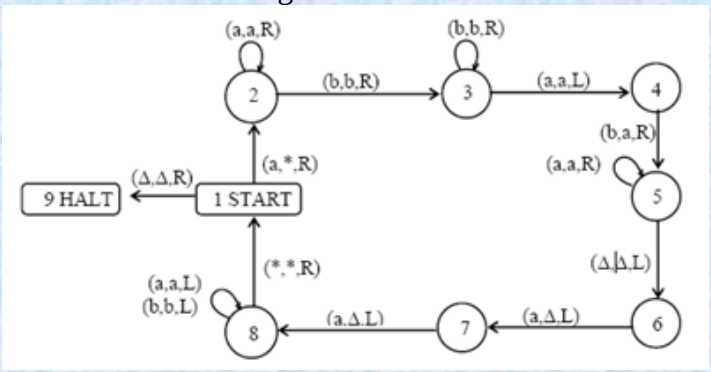
Question No: 19 (Marks: 1) - Please choose one
The process of finding the derivation of the word generated by particular grammar is called _____

►Processing
►**Parsing (Page 136)**
►Programming
►Planning

Question No: 20 (Marks: 1) - Please choose one
The first rule of converting the given "CFG in CNF", is _____

►CNK algorithm
►**CYK algorithm (Page 135) Algorithm 4 (The CYK algorithm)**
►CKY algorithm
►KYC algorithm

Question No: 21 (Marks: 1) - Please choose one
Consider the following TM



►Above TM accepts the non-CFL {a b c}
►**Above TM accepts the non-CFL** $\{a^n\ b^n\ a^n\}$ **(Page 142)**
►Above TM accepts the non-CFL $\{a^n\ b^{n+2}\ a^n\}$

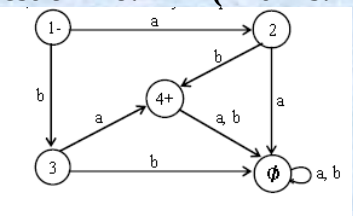
Question No: 22 (Marks: 1) - Please choose one
Alphabet Σ = {a, bc, cc} has number of letters

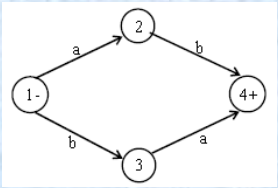
►One
►Two
►**Three**
►Four

دنیا کی سب سے بڑی فتح نفس پر قابو رکھنا ہے

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Question No: 23 (Marks: 1) - Please choose one
If r1 is a regular expression then r1* is a _____
►GTG
►NFA
►FA
►**RE (Page 9)**

Question No: 24 (Marks: 1) - Please choose one
(a)


(b)


► **(a) is FA, (b) is NFA (Page 43)**
► (a) is NFA, (b) is FA
► (a) is TG, (b) is FA
► (a) is TG, (b) is GTG

Question No: 25 (Marks: 1) - Please choose one
We cannot write regular expressions for all _____.

►FA's
►TG's
►NFA's
►**CFG's (Page 97)**

Question No: 26 (Marks: 1) - Please choose one
For every Context Free Grammar (CFG), we can make the corresponding _____.

►FA
►TG
►**PDA** click here for detail
►Regular Grammar

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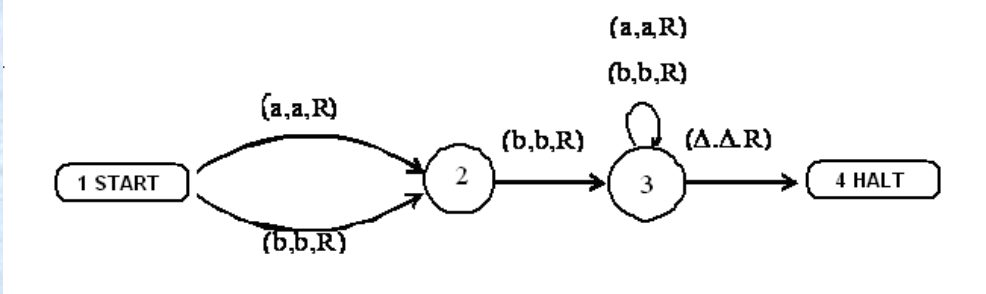
Question No: 27 (Marks: 1) - Please choose one
Pumping Lemma II says that length(x) + length(y) should be _____.

►**Less than number of states (Page 75)**
►Equal to number of states
►Greater than number of states
►Greater than or equal to number of states

Question No: 28 (Marks: 1) - Please choose one
Chomsky normal form (CYK) algorithm was proposed by _____.

►**John cock (Page 135)**
►James Cock
►Daniel I.A.
►John Weiss

Question No: 29 (Marks: 1) - Please choose one



The above machine is a/anTG _____

► Finite Automata
► **Turing machine (Page 141)**
► FA
► TG

Question No: 30 (Marks: 1) - Please choose one
The language of Palindromes defined over an alphabet set {a, b} can be recognized by _____.

►FA
►NFA
►TG
►**PDA (Page 91)**
Hint: - as it is non-regular so its CFG and PDA are possible.

ایماندار کو غصہ دیر سے آتا ہے اور جلدی دور ہو جاتا ہے

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Question No: 31 (Marks: 1) - Please choose one
Which of the following statement(s) is/are true or false?
(1)The Turing Machine is similar to a finite automation but with an unlimited and unrestricted memory.
(2) A Turing machine much more accurate model of a general purpose computer.

►**Statement 1 is true** Click here for detail
►**Statement 2 is true** Click here for detail
►Both statements (1 & 2) are false
►Statements 2 is false

Question No: 32 (Marks: 1) - Please choose one
Which of the following is the first phase of compiler on the basis of functionality?

►Parser
►Lexical analyzer
►**Scanner** Click here for detail
►Interpreter
Hint: - The first phase of a compiler is called lexical analysis (and is also known as a lexical scanner).

Question No: 33 (Marks: 1) - Please choose one
(Σ* - L) represent the _____ of a language L.

►**Complement (Page 66)**
►Kleene's closure
►Union
►intersection

Question No: 34 (Marks: 1) - Please choose one
If we have two transition graphs then their union will be expressed by
►**taking a common start state and joining them by two null transitions (Page 65)**
►just connecting both start states by null transitions
►connecting final state of first TG to the initial state of second TG
►connecting the final state of first TG to the final state of second TG

Question No: 35 (Marks: 1) - Please choose one
_____ and _____ are removed in order to make a CFG in Chomsky Normal Form(CNF).

►Null, nullable productions
►Nullable , unit productions
►**Null, unit productions (Page 102)**
►String of length 0, null

زندگی میں کامیابی کا یہی راز ہے کہ پریشانیوں سے پریشان مت بنو

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Question No: 36 (Marks: 1) - Please choose one
If L1 and L2 are expressed by regular languages then L1 + L2 is also a_____ Language.

► **Regular (Page 10)**
► Ir-regular
► PDA
► Hybrid

Question No: 37 (Marks: 1) - Please choose one
Which of the following is a regular Context Free Grammar:

►**S → abS| baS | ^** **ab(ab+ba)*ba + ba(ab+ba)*ab**
►S → aSb| baS | ^
►S → abS| bSa | ^
►S → aSb| Sa | ^
Hint :- remaining represents palindromes language which is non-regular

Question No: 38 (Marks: 1) - Please choose one
A read state can have _____ outgoing edge/ edges.

►1
►2
►3
►**Any number of (Page 111)**

Question No: 39 (Marks: 1) - Please choose one
Finite Automation (FA) and Nondeterministic Finite Automation (NFA) are equivalent if

►**FA and NFA accept the same language (Page 43)** Also click here for detail
►FA shape is same like an NFA
►FA accept the null string also
►NFA accept the null string also

Question No: 40 (Marks: 1) - Please choose one
_____ is always Deterministic.

►**Finite Automation (Page 25)**
►Transition Graph
►Generalize Transition Graph
►Non-deterministic finite automation

جھوٹ انسان اور ایمان دونوں کا دشمن ہے

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FINALTERM EXAMINATION
Spring 2010
CS402- Theory of Automata (Session - 1)

Question No: 1 (Marks: 1) - Please choose one
If r1 = (aa + bb) and r2 = (a + b) then the language (aa + bb)(a + b) will be generated by

- **(r1)(r2) (Page 10)**
- (r1 + r2)
- (r2)(r1)
- (r1)*

Question No: 2 (Marks: 1) - Please choose one
"One language can be expressed by more than one FA". This statement is _____

- **True (Page 14)**
- False
- Some times true & sometimes false
- None of these

Question No: 3 (Marks: 1) - Please choose one
Who did not invent the Turing machine?

- Alan Turing
- **A. M. Turing (Page 140)**
- Turing
- None of these

Question No: 4 (Marks: 1) - Please choose one
Which statement is true?

- **The tape of turing machine is infinite. (Page 140)**
- The tape of turing machine is finite.
- The tape of turing machine is infinite when the language is regular
- The tape of turing machine is finite when the language is nonregular.

Question No: 5 (Marks: 1) - Please choose one
A regular language:

- **Must be finite (Page 11)**
- Must be infinite
- Can be finite or infinite
- Must be finite and cannot be infinite

عقل مند کہتا ہے میں کچھ نہیں جانتا جبکہ بے وقوف کہتا ہے کہ میں سب کچھ جانتا ہوں

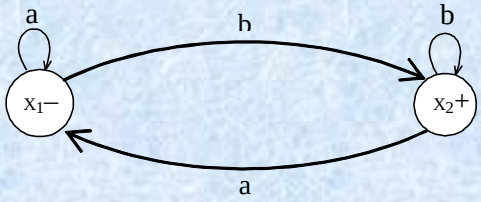
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Question No: 6 (Marks: 1) - Please choose one

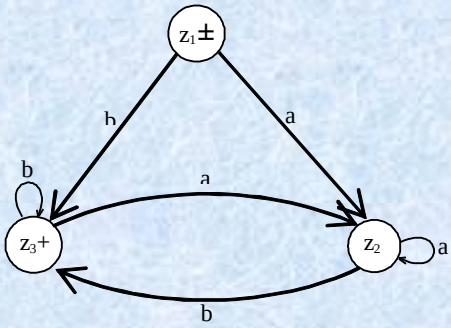
Every regular expression can be expressed as CFG but every CFG cannot be expressed as a regular expression. This statement is:

- ► Depends on the language
- ► None of the given options
- ► **True (Page 97)**
- ► False

Question No: 7 (Marks: 1) - Please choose one



Above given FA corresponds RE r. then FA corresponding to r* will be



This statement is

- ► **True (Page 38)**
- ► False
- ► Depends on language
- ► None of these

Question No: 8 (Marks: 1) - Please choose one

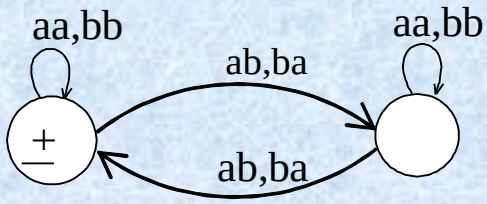
Consider the language L of strings, defined over Σ = {a,b}, ending in a

- ► **There are finite many classes generated by L, so L is regular (Page 76)**
- ► There are infinite many classes generated by L, so L is regular
- ► There are finite many classes generated by L, so L is non-regular
- ► There are infinite many classes generated by L, so L is non-regular

خود کو تمھیں سے بڑھ کر کوئی اچھا مشورہ نہیں دے سکتا

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Question No: 9 (Marks: 1) - Please choose one



Above given TG has _____ RE.

- ► (aa+aa+(ab+ab)(aa+ab)*(ab+ba))*
- ► **(aa+bb+(ab+ba)(aa+bb)*(ab+ba))*** **(Page 22)**
- ► (aa+bb+(ab+ba)(aa+bb)(ab+ba))*
- ► None of these

Question No: 10 (Marks: 1) - Please choose one
The word 'formal' in formal languages means

- ► The symbols used have well defined meaning
- ► They are unnecessary, in reality
- ► **Only the form of the string of symbols is significant** Click here for detail
- ► None of these

Question No: 11 (Marks: 1) - Please choose one
Let A = {0, 1}. The number of possible strings of length 'n' that can be formed by the elements of the set A is

- ► n!
- ► n^2
- ► n^m
- ► **2^n**

Question No: 12 (Marks: 1) - Please choose one
Choose the correct statement.

- ► A Mealy machine generates no language as such
- ► A Moore machine generates no language as such
- ► A Mealy machine has no terminal state
- ► **All of these** click here for detail

Question No: 13 (Marks: 1) - Please choose one
TM is more powerful than FSM because

- ► The tape movement is confined to one direction
- ► It has no finite state control
- ► **It has the capability to remember arbitrary long sequences of input symbols** Click here for detail
- ► None of these

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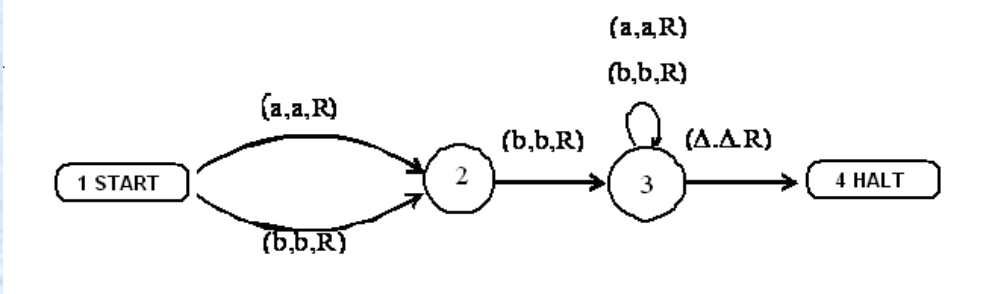
Question No: 14 (Marks: 1) - Please choose one
If L1 and L2 are expressed by regular expressions r1 and r2, respectively then the language expressed by r1 + r2 will be _____

- **Regular (Page 10)**
- Ir-regular
- Can't be decided
- Another Language which is not listed here

Question No: 15 (Marks: 1) - Please choose one
Like TG, a PDA can also be non-deterministic

- **True (Page 111)**
- False

Question No: 16 (Marks: 1) - Please choose one



The above machine is a/an _____

- Finite Automata
- **Turing machine (Page 141) rep**
- FA
- TG

Question No: 17 (Marks: 1) - Please choose one
The language of all words (made up of a's and b's) with at least two a's can not be described by the regular expression.

- a(a+b)*a(a+b)*(a+b)*ab*
- (a+b)* ab* a(a+b)*
- b*ab* a(a+b)*
- **none of these**

a^nb^n {where n>0} is the language will at least one a and b and cannot be described by RE.

Question No: 18 (Marks: 1) - Please choose one
In FA, if one enters in a specific state but there is no way to leave it, then that specific state is called

- Dead State
- Waste Basket
- Davey John Locker
- **All of these (Page 17)**

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Question No: 19 (Marks: 1) - Please choose one
If L is a regular language then, L^c is also a _____ language.

► **Regular (Page 66) rep**
► Non-regular
► Regular but finite
► None of the given

Question No: 20 (Marks: 1) - Please choose one
In CFG, the symbols that can't be replaced by anything are called___

► **Terminal (Page 87) rep**
► Non-Terminal
► Production
► All of given

Question No: 21 (Marks: 1) - Please choose one
Which of the following is NOT a regular language?

► String of 0's whose length is a perfect squere
► Set of all palindromes made up of 0's and 1's
► String of 0's whose length is a prime number
► **All of the given options** Click here for detail

Question No: 22 (Marks: 1) - Please choose one
Choose the incorrect (FALSE) statement.

► A Mealy machine generates no language as such
► A Mealy machine has no terminal state
► **For a given input string, length of the output string generated by a Moore machine is not more than the length of the output string generated by that of a Mealy machine** click here for detail
► All of these

Question No: 23 (Marks: 1) - Please choose one
Pumping lemma is generally used to prove that:

► A given language is infinite
► **A given language is not regular** Click here for detail **rep**
► Whether two given regular expressions of a regular language are equivalent or not
► None of these

Question No: 24 (Marks: 1) - Please choose one
Which of the following is a regular language?

► **String of odd number of zeroes** Click here for detail **rep**
► Set of all palindromes made up of 0's and 1's
► String of 0's whose length is a prime number
► All of these

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Question No: 25 (Marks: 1) - Please choose one
Choose the incorrect statement:
► (a+b)*aa(a+b)* generates Regular language.
► **A language consisting of all strings over ∑={a,b} having equal number of a's and b's is a regular language**
► Every language that can be expressed by FA can also be expressed by RE
► None of these

Question No: 26 (Marks: 1) - Please choose one
Left hand side of a production in CFG consists of:
► One terminal
► More than one terminal
► **One non-terminal (Page 87)**
► Terminals and non-terminals

FINALTERM EXAMINATION
SPRING 2007

Question No: 1 (Marks: 1) - Please choose one
PDA is only used to represent a regular language.
► True
► **False** Click here for detail

Question No: 2 (Marks: 1) - Please choose one
If L is a regular language then LC is also a regular language.
► **True (Page 66) rep**
► False

Question No: 3 (Marks: 1) - Please choose one
A production of the form non-terminal ⇾ string of two non-terminal is called a live Production.
► **True (Page 127)**
► False

Question No: 4 (Marks: 1) - Please choose one
We can find a CFG corresponding to a DFA.
► **True (Page 97)**
► False

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Question No: 5 (Marks: 1) - Please choose one
START, READ, HERE and ACCEPTS are conversions of the machine
► **True (Page 122)**
► False

Question No: 6 (Marks: 1) - Please choose one
A CFG is said to be ambiguous if there exists at least one word of its language that can be generated by different production trees
► **True (Page 95)**
► False

Question No: 7 (Marks: 1) - Please choose one
Syntax tree or Generation tree or Derivation tree are same tree
► **True (Page 92)**
► False

Question No: 8 (Marks: 1) - Please choose one
The symbols that cannot be replaced by anything are called terminals
► **True (Page 87) rep**
► False

Question No: 9 (Marks: 1) - Please choose one
The production of the form non-terminal ⇾ one non-terminal is called unit production
► **True (Page 100)**
► False

Question No: 10 (Marks: 1) - Please choose one
DFA and PDA are equal in power.
► True
► **False (Page 105)**

FINALTERM EXAMINATION
Spring 2006
CS402- Theory of Automata

Question No. 1
A production of the form non-terminal ⇾ non-terminal is called a dead Production.
True
False (Page 127)

جو شخص ناکامیوں سے ڈر کر بھاگتا ہے کامیابی اُس سے ڈر کر بھاگتی ہے

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Question No. 2
Semi-word is a string having some terminals and one non-terminal at the right of string.
True (Page 97)
False

Question No. 3
Two FAs are equivalent if they have same no. of states.
True (Page 15)
False

Question No. 4
There exist exactly two different derivations in an ambiguous CFG for a word.
True (Page 93)
False

Question No. 6
Regular languages are closed under Union, Concatenation and Kleene star.
True (Page 10)
False

Question No. 7
CFG may also represent a regular language.
True (Page 97)
False

Question No. 9 Marks : 1
PDA is stronger than FA.
True (Page 105)
False

FINALTERM EXAMINATION
Spring 2005
CS402- Theory of Automata

Question No. 1
A Total Language Tree has
All languages over Σ
All strings over Σ (Page 96)
All words of all languages over Σ
All words of one language over Σ

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Question No. 2
What Turing Machine does not have?
Stack
Tape
Head
Word
Turing machine has stack but insertion and deletion can be done from both sides. Tape and head to.

Question No. 3
CFG given S ⇗ bS|Sb|aa represents language
b*aa
aab*
b*aab*
b*(aa)*b*

Question No. 4
A Language that is finite but not regular
Λ
(a+b)*
Φ (not sure)
All strings of a's in Σ = {a,.b}

جو لوگوں کے سامنے فخر کرتا ہے وہ لوگوں کی نظروں سے گر جاتا ہے

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CS402 – Quiz No.3

Question # 1 of 10 (Total Marks: 1) **Select correct option:**
The values of input (say a & b) does not remain same in one cycle due to
NAND gate
Click plus
OR gate
NOT gate

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
Set of all palindromes over {a,b}is regular

True
False (Page 74)

Question # 3 of 10 (Total Marks: 1) **Select correct option:**
In CFG, the symbols that cannot be replaced by anything are called

Terminals (Page 87) rep
Non terminals
Productions
None of the given options

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
a^n b^n generates the …… language

regular
non regular
EQUAL and non regular (Page 71)
EQUAL and regular

Question # 5 of 10 (Total Marks: 1) **Select correct option:**
The grammatical rules which involves meaning of words are called:

Semantic (Page 87)
Sytactics
Alphabets
None of the given options

عقل مند اپنے عیب خود دیکھتا ہے اور بیوقوفوں کے عیب دنیا دیکھتی ہے

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Question # 6 of 10 (Total Marks: 1) **Select correct option:**
The reverse of the string sbfsbb over { sb, f, b }

bbsfbs
bsbfsb
sbbfsb
bsfbsb

Question # 7 of 10 (Total Marks: 1) **Select correct option:**
If an FA has N state then it must accept the word of length

N-1
N+1
N+2N

Question # 8 of 10 (Total Marks: 1) **Select correct option:**
Two languages are said to belong to same class if they end in the same state when they run over an FA, that state

Must be final state
May be final state or not (Page 75)
May be start or not
None of the given options

Question # 9 of 10 (Total Marks: 1) **Select correct option:**
In pref(Q in R) Q is …… to (than) R

Equal
Not Equal (Page 79)
Greater
Smaller

Question # 10 of 10 (Total Marks: 1) **Select correct option:**
According to Myhill Nerode theorem, if L generates finite no. of classes then L is.....

Finite
Infinite
Regular (Page 76)
Non Regular

عقل مند آدمی اس وقت تک نہیں بولتا جب تک خاموشی نہیں ہو جاتی

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Question # 1 of 10 (Total Marks: 1) **Select correct option:**
If the intersection of two regular languages is regular then the complement of the intersection of these two languages is also regular

True (Page 68)
False

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
In pumping lemma theorem (x y^n z) the range of n is

n=1,2,3,4..... (Page 74)
n=0,1,2,3,4....
n=-3,-2,-1,0,1,2,3,4.....
n=-3,-2,-1,1,2,3,4.....

Question # 3 of 10 (Total Marks: 1) **Select correct option:**
The complement of a regular language is also a regular

True rep
False

CS402 – Quiz No.3

Question # 1 of 10 (Total Marks: 1) **Select correct option:**
For a non regular language there exist …… FA

One
At least one
At most one
No (Page 71)

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
The strings or words which do not belong to a language is called……. of that language

Intersection
Union
Complement (Page 66)
Quotient

انسان دکھ نہیں دیتے بلکہ انسانوں سے وابستہ امیدیں دکھ دیتی ہیں

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Question # 3 of 10 (Total Marks: 1) **Select correct option:**
A non regular language can be represented by
RE
FA
TG
None of the given options (Page 71)

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
For language L defined over {a, b},then L partitions {a, b}* into …… classes

Infinite
Finite
Distinct (Page 75)
Non distinct

Question # 5 of 10 (Total Marks: 1) **Select correct option:**
If an FA accept a word then there must exist a path from

Initial to final state (Page 81)
Initial to each state
Initial to each state but not to final state
Initial to final state by traversing each state

Question # 6 of 10 (Total Marks: 1) **Select correct option:**
Does the empty string match the regular expression |y+a|?

Yes
No (Page 3)

Question # 7 of 10 (Total Marks: 1) **Select correct option:**
If an FA already accepts the language expressed by the closure of certain RE, then the given FA is the required FA.

True (Page 37)
False

Question # 8 of 10 (Total Marks: 1) **Select correct option:**
Which of the following statement is true about NFA with Null String?

Infinite states
Infinite set of letters
Infinite set of transitions
Transition of null string is allowed at any stage (Page 71)

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Question # 9 of 10 (Total Marks: 1) **Select correct option:**
If R is a regular language and L is some language, and L U R is a regular language, then L must be a regular language.

True (page 10)
False

Question # 10 of 10 (Total Marks: 1) **Select correct option:**
FA corresponding to an NFA can be built by introducing an empty state for a letter having

no transition at certain state (Page 43)
one transition at certain state
two transition at certain state
more than two transitions at certain state

Question # 1 of 10 (Total Marks: 1) **Select correct option:**
Let FA3 be an FA corresponding to FA1FA2, then the initial state of FA3 must correspond to the initial state of

FA1 only (Page 35)
FA2 only
FA1 or FA2
FA1 and FA2

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
(a* + b*)* = (a + b)* this expression is _____
True
False

Question # 3 of 10 (Total Marks: 1) **Select correct option:**
If S = { x }, then S* will be
{x,xx,xxx,xxxx,…}
{^ ,x,xx,xxx,xxxx,…} (Page 10)

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
The states in which there is no way to leave after entry are called
Davey John Lockers
Dead States
Waste Baskets
All of the given options (Page 17)

خوبصورتی علم وادب سے ہوتی ہے لباس و حسن سے نہیں

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Question # 5 of 10 (Total Marks: 1) **Select correct option:**
If S = {ab, bb}, then S* will not contain
Abbbab
Bbba
ababbb
bbbbab

Question # 6 of 10 (Total Marks: 1) **Select correct option:**
According to theory of automata there are _____ types of languages

1
2 (Page 3)
3
4

Question # 7 of 10 (Total Marks: 1) **Select correct option:**
What do automata mean?
Something done manually
Something done automatically (Page 3)

Question # 8 of 10 (Total Marks: 1) **Select correct option:**
What is false about the term alphabet?
It is a finite set of symbols.
It is usually denoted by Greek letter sigma
It can be an empty set. (Page 3)
Strings are made up of its elements

Question # 9 of 10 (Total Marks: 1) **Select correct option:**
Formal is also known as _____
Syntactic language (page 3)
Semantic language
Informal language
Nsone of these

Question # 10 of 10 (Total Marks: 1) **Select correct option:**
Following are types of languages

Formal Languages (Syntactic languages)
Informal Languages (Semantic languages)
Both (Page 3)
None of above

بہترین تجربہ وہ ہے جس سے نصیحت حاصل ہو

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CS402 – Quiz No.4

Question # 1 of 10 (Total Marks: 1) **Select correct option:**
Consider the following production (of a CFG): S->XYZ Here _____ is left most nonterminal in working string. Note: S, X, Y and Z are all nonterminals
S
X
Y
Z

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
A PDA is called nondeterministic PDA if _____

There are more than one outgoing edges at READ or POP states with one label (Page 111)
There are more than one PUSH states
There are mroe than one POP states
All of the given options

Question # 3 of 10 (Total Marks: 1) **Select correct option:**
A PDA consists of the following:

An alphabet (Sigma) of input letters.
An input TAPE with infinite many locations in one direction
One START state with only one out-edge and no in-edge
All of the given options (Page 105)

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
The CFG S --> aSa | bSb | a | b | ^ represents the language

EVEN-EVEN
PALINDROM (Page 91)
EQUAL
ODD-ODD

Question # 5 of 10 (Total Marks: 1) **Select correct option:**
Halt states are

Start and Accept
Accept and Reject (Page 105)
Start and Reject
Read and Reject

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Question # 6 of 10 (Total Marks: 1) **Select correct option:**
Choice of path can be determined by left most derivation of the string belonging to CFL at…… state

Accept (Page 104)
Reject
Push
POP

Question # 7 of 10 (Total Marks: 1) **Select correct option:**
The unit and null productions can be deleted from a CFG

True (Page 99-100)
False

Question # 8 of 10 (Total Marks: 1) **Select correct option:**
Identify the TRUE statement about following CFG:
S -> SB|AB
A -> CC
B -> b
C -> a

The given CFG has 8 Nonterminals
The given CFG has 8 Terminals
The given CFG is in CNF (Page 101)
The given CFG is not in CNF

Question # 9 of 10 (Total Marks: 1) **Select correct option:**
The structure given below is called _____ S -> aA|bB A -> aS|a B -> bS|b

RE
TG
CFG (Page 87)
PDA

Question # 10 of 10 (Total Marks: 1) **Select correct option:**
Which of the following states is not part of PDA

START
ACCEPT
WRITE (Page 107)
REJECT

تم اچھا کرو زمانہ تم کو برا سمجھے یہ اس سے بہتر ہے کہ تم برا کرو اور زمانہ تم کو اچھا سمجھے

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CS402 – Quiz No.4

Question # 1 of 10 (Total Marks: 1) **Select correct option:**
The production of the form: nonterminal --> one nonterminal is called the _____

Unit production (Page 100)
NULL production
Terminal production
Non Terminal production

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
A _____ is the one for which every input string has a unique path through the machine.

Deterministic PDA (Page 111)
nondeterministic PDA
PUSHDOWN store
Input Tape

Question # 3 of 10 (Total Marks: 1) **Select correct option:**
In the null production N --> ^ , N is a

Terminal
Non terminal (Page 99)
Word
None of the given options

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
The major problem in the earliest computers was

To store the contents in the registers
To display mathematical formulae (Page 87)
To load the contents from the registers
To calculate the mathematical formula

Question # 5 of 10 (Total Marks: 1) **Select correct option:**
In polish notation, (o-o-o) is the abbreviation of……?

Operand - Operator – Operand
Operand - Operand- Operator
Operator -Operand – Operand (Page 94)
Operand -Operand – Operand

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Question # 6 of 10 (Total Marks: 1) **Select correct option:**
The CFG is said to be ambiguous if there exist at least one word of its language that can be generated by the …… production trees

One
Two
More than one (Page 95)
At most one

Question # 7 of 10 (Total Marks: 1) **Select correct option:**
The input string is placed, before it runs, in

Stack
Memory
Tape (Page 105)
Ram

Question # 8 of 10 (Total Marks: 1) **Select correct option:**
The production S --> SS | a | b | ^ can be expressed by RE

(a+b)+
(a+b)
(a+b)* (Page 88)
(ab)*

Question # 9 of 10 (Total Marks: 1) **Select correct option:**
The locations into which we put the input letters on "Input Tap" are called _____

words
alphabets
cells (Page 105)
elements

Question # 10 of 10 (Total Marks: 1) **Select correct option:**
"CFG" stands for _____

Context Free Graph
Context Free Grammer (Page 87)
Context Finite Graph
Context Finite Grammer

بدصورت چہرہ بدصورت دماغ سے بہتر ہے

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Question # 1 of 10 (Total Marks: 1) **Select correct option:**
In a CFG the nonterminal that occurs first from the left in the working string, is said to be _____

Least Significant nonterminal
Most Significant nonterminal
Left most nonterminal (Page 103)
Left most derivate

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
The unit production is

Terminal --> Terminal
Terminal --> Non Terminal
Non terminal --> Terminal
Non terminal --> Non Terminal (Page 100)

Question # 3 of 10 (Total Marks: 1) **Select correct option:**
A _____ operator adds a new letter at the top of STACK

PUSH (Page 107)
POP
READ
APPEND

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
PDA stands for _____

Push and Drop Automaton
Pop and Drop Automaton
Push Down Automaton (Page 112)
None of given options

Question # 5 of 10 (Total Marks: 1) **Select correct option:**
The production of the form: Nonterminal-> ^ is said to be _____ production

NULL (Page 99)
UNIT
Chomsky form production
None of the given options

عقل مند آدمی اس وقت تک نہیں بولتا جب تک خاموشی نہیں ہو جاتی

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Question # 6 of 10 (Total Marks: 1) **Select correct option:**
If a CFG has a null production, then it is _____

Posiible to construct another CFG without null production accepting the same language with the exception of the word ^ (Page 99)
Not possible to construct another CFG without null production accepting the same language with the exception of the word ^
Called NULL CFG
Called Chmosky Normal Form (CNF)

Question # 7 of 10 (Total Marks: 1) **Select correct option:**
In a STACK:

The element PUSHed first is POPed first
The element PUSHed first is POPed in the last (Page 107 concept)
The element PUSHed in last is POPed in last
None of given options

Question # 8 of 10 (Total Marks: 1) **Select correct option:**
Kleene star closure can be defined
Over any set of string (Page 7)
Over specific type of string

Question # 9 of 10 (Total Marks: 1) **Select correct option:**
While finding RE corresponding to TG, we connect the new start state to the old start state by the transition labeled by
A
B
null string (Page 26)
None of the given options

انسان کے لئے بری صحبت سے بڑھ کر بری کوئی چیز نہیں

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Some More Quizzes

Question # 1 of 10 (Total Marks: 1) **Select correct option:**
For a given input, it provides the compliment of Boolean AND output.

NAND box (NOT AND) **(Page 63)**
DELAY box
OR box
AND box

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
It delays the transmission of signal along the wire by one step (clock pulse).

NAND box (NOT AND)
DELAY box **(Page 63)**
OR box
AND box

Question # 3 of 10 (Total Marks: 1) **Select correct option:**
For the given input, it provides the Boolean OR output

NAND box (NOT AND)
DELAY box
OR box **(Page 63)**
AND box

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
For the given input, AND box provides the Boolean AND output.
True **(Page 63)**
False

Question # 5 of 10 (Total Marks: 1) **Select correct option:**
The current in the wire is indicated by 1 and 0 indicates the absence of the current.
True **(Page 63)**
False

Question # 6 of 10 (Total Marks: 1) **Select correct option:**
Any language that can not be expressed by a RE is said to be regular language.
True
False **(Page 71)**

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Question # 7 of 10 (Total Marks: 1) **Select correct option:**
If L1 and L2 are regular languages_____ is/are also regular language(s).

L1 + L2
L1L2
L1*
All of above (Page 10)

Question # 8 of 10 (Total Marks: 1) **Select correct option:**
Let L be a language defined over an alphabet Σ, then the language of strings, defined over Σ, not belonging to L, is called Complement of the language L, denoted by Lc or L'.
True (Page 66)
False

Question # 9 of 10 (Total Marks: 1) **Select correct option:**
To describe the complement of a language, it is very important to describe the ---------- of that language over which the language is defined.

Alphabet (Page 66)
Regular Expression
String
Word

Question # 10 of 10 (Total Marks: 1) **Select correct option:**
For a certain language L, the complement of Lc is the given language L *i.e.* (Lc)c = Lc
True
False (Page 66)

Question # 1 of 10 (Total Marks: 1) **Select correct option:**
If L is a regular language then, --------- is also a regular language.
Lm
Ls
Lx
Lc (Page 66)

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
Converting each of the final states of F to non-final states and old non-final states of F to final states, FA thus obtained will reject every string belonging to L and will accept every string, defined over Σ, not belonging to L. is called
Transition Graph of L
Regular expression of L
Complement of L (Page 66)
Finite Automata of L

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Question # 3 of 10 (Total Marks: 1) **Select correct option:**
If L1 and L2 are two regular languages, then L1 U L2 is not a regular.
True
False (Page 65)

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
De-Morgan's law for sets is expressed by,
$(L_1^c \cap L_2^c)^c = L_1^c \cap L_2^c$
$(L_1^c \cap L_2^c)^c = L_1^c \cap L_2^c$
$(L_1^c \cap L_2^c)^c = L_1 \cap L_2$
$(L_1^c \cap L_2^c)^c = L_1 \cup L_2$ **CORRECT (page 68)**

Question # 5 of 10 (Total Marks: 1) **Select correct option:**
If L1 and L2 are regular languages, then these can be expressed by the corresponding FAs.
True (Page 68)
False

Question # 6 of 10 (Total Marks: 1) **Select correct option:**
L= language of words containing even number of a's. Regular Expression is

(a+b)*aa(a+b)*
(b+ab*a)* (Page 68)
a+bb*aab*a
(a+b)*ab(a+b)*

Question # 7 of 10 (Total Marks: 1) **Select correct option:**
The regular expression defining the language $L_1 \cap L_2$ can be obtained, converting and reducing the previous ---------- into a ---------- as after eliminating states.

GTG, TG
FA, GTG (Page 69)
FA, TG
TG, RE

Question # 8 of 10 (Total Marks: 1) **Select correct option:**
The language that can be expressed by any regular expression is called a Non regular language.
True
False (Page 71)

خاموشی غصے کا بہترین علاج ہے

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Question # 9 of 10 (Total Marks: 1) **Select correct option:**
The languages ---------- are the examples of non regular languages.

PALINDROME and PRIME (Page 71)
PALINDROME and EVEN-EVEN
EVEN-EVEN and PRIME
FACTORIAL and SQURE

Question # 10 of 10 (Total Marks: 1) **Select correct option:**
Let L be any infinite regular language, defined over an alphabet Σ then there exist three strings x, y and z belonging to Σ* such that all the strings of the form xy^nz for n=1,2,3, … are the words in L. called.

Complement of L
Pumping Lemma (Page 72)
Kleene's theorem
None in given

Question # 1 of 10 (Total Marks: 1) **Select correct option:**
Languages are proved to be regular or non regular using pumping lemma.
True (Page 74)
False

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
---------- is obviously infinite language.
EQUAL-EQUAL
EVEN-EVEN
PALINDROME (Page 75)
FACTORIAL

Question # 3 of 10 (Total Marks: 1) **Select correct option:**
If, two strings x and y, defined over Σ, are run over an FA accepting the language L, then x and y are said to belong to the same class if they end in the same state, no matter that state is final or not.
True (Page 75)
False

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
Myhill Nerode theorem is consisting of the followings,

L partitions Σ* into distinct classes.
If L is regular then, L generates finite number of classes.
If L generates finite number of classes then L is regular.
All of above (Page 75)

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Question # 5 of 10 (Total Marks: 1) **Select correct option:**
The language Q is said to be quotient of two regular languages P and R, denoted by--- if PQ=R.
R=Q/P
Q=R/P (Page 78)
Q=P/R
P=R/Q

Question # 6 of 10 (Total Marks: 1) **Select correct option:**
If two languages R and Q are given, then the prefixes of Q in R denoted by Pref(Q in R).
True (Page 78)
False

Question # 7 of 10 (Total Marks: 1) **Select correct option:**
Let Q = {aa, abaaabb, bbaaaaa, bbbbbbbbbb} and R = {b, bbbb, bbbaaa, bbbaaaaa} Pref (Q in R) is equal to,

{b,bbba,bbbaaa} (Page 78)
{b,bba,bbaaa}
{ab,bba,bbbaa}
{b,bba,bbba}

Question # 8 of 10 (Total Marks: 1) **Select correct option:**
If R is regular language and Q is any language (regular/ non regular), then Pref (Q in R) is ---------.

Non-regular
Equal
Regular (Page 79)
Infinite

Question # 9 of 10 (Total Marks: 1) **Select correct option:**
_____ states are called the halt states.
ACCEPT and REJECT (Page 105)
ACCEPT and READ
ACCEPT AND START
ACCEPT AND WRITE

Question # 10 of 10 (Total Marks: 1) **Select correct option:**
The part of an FA, where the input string is placed before it is run, is called _____

State
Transition
Input Tape (Page 105)
Output Tape

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Question # 1 of 10 (Total Marks: 1) **Select correct option:**
In new format of an FA (discussed in lecture 37), This state is like dead-end non final state

ACCEPT
REJECT (Page 105)
STATR
READ

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
Between the two consecutive joints on a path

One character can be pushed and one character can be popped
Any no. of characters can be pushed and one character can be popped (Page 122)
One character can be pushed and any no. of characters can be popped
Any no. of characters can be pushed and any no. of characters can be popped

Question # 3 of 10 (Total Marks: 1) **Select correct option:**
The PDA is called non-deterministic PDA when there are more than one out going edges from…… state

START or READ
POP or REJECT
READ or POP (Page 111)
PUSH or POP

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
Identify the TRUE statement:
A PDA is non-deterministic, if there are more than one READ states in PDA
A PDA is never non-deterministic
Like TG, A PDA can also be non-deterministic (Page 111)
A PDA is non-deterministic, if there are more than one REJECT states in PDA

Question # 5 of 10 (Total Marks: 1) **Select correct option:**
There is a problem in deciding whether a state of FA should be marked or not when the language Q is infinite.
True (Page 79)
False

Question # 6 of 10 (Total Marks: 1) **Select correct option:**
If an effectively solvable problem has answered in yes or no, then this solution is called ---------

Decision procedure (Page 80)
Decision method
Decision problem
Decision making

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Question # 7 of 10 (Total Marks: 1) **Select correct option:**
The following problem(s) ---------- is/are called decidable problem(s).

The two regular expressions define the same language
The two FAs are equivalent
Both a and b (Page 80)
None of given

Question # 8 of 10 (Total Marks: 1) **Select correct option:**
To examine whether a certain FA accepts any words, it is required to seek the paths from ------- state.

Final to initial
Final to final
Initial to final (Page 81)
Initial to initial

Question # 9 of 10 (Total Marks: 1) **Select correct option:**
The high level language is converted into assembly language codes by a program called compiler.

TRUE (Page 87)
FALSE

Question # 10 of 10 (Total Marks: 1) **Select correct option:**
Grammatical rules which involve the meaning of words are called ----------
Semantics (Page 87)
Syntactic
Both a and b
None of given

Question # 1 of 10 (Total Marks: 1) **Select correct option:**
Grammatical rules which do not involve the meaning of words are called ----------
Semantics
Syntactic (Page 87)
Both a and b
None of given

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
The symbols that must be replaced by other things are called _____

Productions
Terminals
Non-terminals (Page 87)
None of given

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Question # 3 of 10 (Total Marks: 1) **Select correct option:**
The grammatical rules are often called_____

Productions (Page 87)
Terminals
Non-terminals
None of given

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
The terminals are designated by _____ letters, while the non-terminals are designated by _____ letters.
Capital, bold
Small, capital (Page 87)
Capital, small
Small, bold

Question # 5 of 10 (Total Marks: 1) **Select correct option:**
The language generated by _____ is called Context Free Language (CFL).
FA
TG
CFG (Page 87)
TGT

Question # 6 of 10 (Total Marks: 1) **Select correct option:**
$\Sigma = \{a,b\}$ Productions $S \rightarrow XaaX$
$X \rightarrow aX$
$X \rightarrow bX$
$X \rightarrow \Lambda$
This grammar defines the language expressed by_____

(a+b)*aa(a+b)* (Page 89)
(a+b)*a(a+b)*a
(a+b)*aa(a+b)*aa
(a+b)*aba+b)*

Question # 7 of 10 (Total Marks: 1) **Select correct option:**
$S \rightarrow aXb|bXa$
$X \rightarrow aX|bX|\Lambda$
The given CFG generates the language in English _____

Beginning and ending in different letters (Page 91)
Beginning and ending in same letter
Having even-even language
None of given

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Question # 8 of 10 (Total Marks: 1) **Select correct option:**
The CFG is not said to be ambiguous if there exists atleast one word of its language that can be generated by the different production trees,

TRUE
FALSE (Page 95)

Question # 9 of 10 (Total Marks: 1) **Select correct option:**
The language generated by that CFG is regular if _____

No terminal → semi word
No terminal → word
Both a and b (Page 97)
None of given

Question # 10 of 10 (Total Marks: 1) **Select correct option:**
The production of the form no terminal → Λ is said to be null production.

TRUE (Page 99)
FALSE

Question # 1 of 10 (Total Marks: 1) **Select correct option:**
CNF is stands for

Context Normal Form
Complete Normal Form
Chomsky Normal Form (Page 102)
Compared Null Form

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
Proof(Kleene's Theorem Part II)
If a TG has more than one start states, then

Introduce the new start state (Page 26)
Eliminate the old start state
Replace the old start state with final state
Replace the old final state with new start state

Question # 3 of 10 (Total Marks: 1) **Select correct option:**
Which of the following regular expression represents same language?
a. (a+ab)*
b. (ba+a)*
c. a*(aa*b)*
d. (a*b*)* (a+b)*a(a+b)*b(a+b)*+ (a+b)*b(a+b)*a(a+b)*.

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{ x}*, { x}+, {a+b}*

Select correct option:
a and b (correct)
a and c
c and d

Question # 4 of 10 (Total Marks: 1) Select correct option:
Let FA3 be an FA corresponding to FA1+FA2, then the initial state of FA3 must correspond to the initial state of

FA1 only
FA2 only
FA1 or FA2 (Page 32)
FA1 and FA2

Question # 5 of 10 (Total Marks: 1) Select correct option:
Which of the following statement is NOT true about TG?

There exists exactly one path for certain string (Page 19)
There may exist more than one paths for certain string
There may exist no path for certain string
There may be no final state

Question # 6 of 10 (Total Marks: 1) Select correct option:
Kleene's theorem states

All representations of a regular language are equivalent.
All representations of a context free language are equivalent.
All representations of a recursive language are equivalent
Finite Automata are less powerful than Pushdown Automata. (Page 105)

Question # 7 of 10 (Total Marks: 1) Select correct option:
A language accepted by an FA is also accepted by

TG only
GTG only
RE only
All of the given (Page 25)



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Quiz No. 4

Question # 1of 10 (Total Marks: 1) **Select correct option:**
Consider the Following CFG: (NOTE: ^ means NULL) S->Xa X->aX|bX|^
above given CFG can be represented by RE _____

a*b*
a*b*a
(a+b)*a
a(a+b)*a

Question # 2 of 10 (Total Marks: 1) **Select correct option:**
Identify FALSE statement:

Every Regular Expression be expressed by CFG and every CFG can be expressed by a Regular Expression (Page 97)
Every regular expression can be expressed as CFG but every CFG cannot be expressed as a regular expression.
For a PDA, there exists a CFG, that represents the same language as represented by PDA.
None of the given options

Question # 3 of 10 (Total Marks: 1) **Select correct option:**
Null production is a

Word
String
Terminal
All of the given options

Question # 4 of 10 (Total Marks: 1) **Select correct option:**
In nondeterministic PDA a string is supposed to be accepted, if there exists at least one path traced by the string, leading to _____ state.

ACCEPT (Page 111)
REJECT
START
READ

افضل انسان وہ ہے جو اپنی اصلاح کی کوشش کرتا ہے

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Question # 5 of 10 (Total Marks: 1) **Select correct option:**
The CFG which generates the regular language is called

Regular expression
Finite Automata
Regular grammar (Page 97)
None of the given options

Question # 6 of 10 (Total Marks: 1) **Select correct option:**
If a CFG has a null production, then it is possible to construct another CFG accepting the same language without null production

TRUE (Page 99)
FALSE

CS402 – Quiz No.2 (15 Jun 2013)

Question # 1 of 10 (Total Marks: 1) Select correct option
In large FA with thousands of states and millions of directed edges, without an effective procedure it is _____ to find a path from initial to final state.
Always easy
Impossible (Page 81)
may be good
always impossible

Question # 2 of 10 (Total Marks: 1) Select correct option
If there is no final state of two FAs then their _____ also have no _____ state
initial, union
final, union
union,final (Page 83)
union, initial

Question # 3 of 10 (Total Marks: 1) Select correct option
Set of all palindromes over {a,b} is:
Regular
Regular and finite
Regular and infinite
Non-regular (Page 71)

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Question # 4 of 10 (Total Marks: 1) Select correct option
In the context of Myhill Nerode theorem, for even-even language sigma star can be partitioned into _____ number of classes.

3
4 (Page 77)
5
6

Question # 5 of 10 (Total Marks: 1) Select correct option
The product of two regular languages is _____.

Regular (Page 78)
infinite
non-regular
closure of a regular language

Question # 6 of 10 (Total Marks: 1) Select correct option
If the FA has N states, then test the words of length less than N. If no word is accepted by this FA, then it will _____ word/words.

accept all
accept no (Page 85) rep
accept some
reject no

Question # 7 of 10 (Total Marks: 1) Select correct option
An FA has same initial and _____ state, then it means that it has no _____ state.

initial, final
final, initial
initial, initial
none of the given options

Question # 8 of 10 (Total Marks: 1) Select correct option
A problem that has decision procedure is called _____ problem.

Regular language
un-decidable
Infinite
Decidable (Page 80)

اطمینان قلب چاہتے ہو تو حسد سے دور رہو

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Question # 9 of 10 (Total Marks: 1) Select correct option
For a machine with N number of states, the total number of strings to be tested, defined over an alphabet of m letters, is_____.
Select correct option:

Nm +Nm+1+ N m+2 +… + N2m-1
m^N + m^N+1 + m^N + 2 +… +m2N-1 (Page 86)
Nm
mN

Question # 10 of 10 (Total Marks: 1) Select correct option
If (L1 ∩ L2c) ∪ (L1c ∩ L2) is regular language that accepts the words which are in L1 but not in L2 or else in L2 but not in L1 . The corresponding FA cannot accept any word which is in _____ L1 and L2.
Not both
Both (Page 80)
At least in one
None of the given options

Question # 1 of 10 (Total Marks: 1) Select correct option
While determining regular expression for a given FA, it is _____ to write its regular expression.

Always possible easily
Sometime impossible (Page 80)
always impossible
None of the given options

Question # 2 of 10 (Total Marks: 1) Select correct option
Incase of Myhill Nerode theorem, if a language L partitions sigma star into distinct classes and L is also regular then L generates_____ number of classes.

infinite
specified
finite (Page 75)
odd

ہر چیز کی ایک پہچان ہوتی ہے اور عقلمند کی پہچان غور و فکر کرنا ہے اور غور و فکر کی پہچان خاموشی ہے

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بِسْمِ اللّٰهِ الرَّحْمٰنِ الرَّحِيْمِ

In the Name of Allāh, the Most Gracious, the Most Merciful

Paper Pattern

MCQS 40 each 1 mark
Short 4 each 2 marks
Short 4 each 3 marks
long 4 each 5 marks

Question No : 1 of 52 Marks: 1 (Budgeted Time 1 Min)

Which statement is true?

Answer (Please select your correct option)

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- ● All words are strings
- ○ All strings are words
- ○ Both are always same
- ○ All letter are string

Lec 1,page:4

Made by: Waqar Siddhu

Question No : 2 of 52 Marks: 1 (Budgeted Time 1 Min)

If L1 and L2 are expressed by r1 and r2, respectively then the language expressed by (r1)(r2) will be _____

Answer (Please select your correct option)

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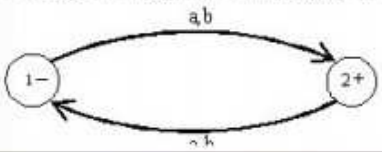
- ● Regular
- ○ Ir-regular
- ○ Undermined
- ○ Such language does not exist

Lec:4,page:10

Made by: Waqar Siddhu

Question No : 3 of 52 Marks: 1 (Budgeted Time 1 Min)

Below given FA generates the language having strings of _____



Answer (Please select your correct option)

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- ● ODD length

conceptual RE=(a+b)((a+b)(a+b))*

- ○ EVEN length
- ○ Equal number of a's and b's
- ○ Odd number of a's and b's

Made by: Waqar Siddhu

Question No : 4 of 52 Marks: 1 (Budgeted Time 1 Min)

If an FA accepts no string, then language is denoted by _____

Answer (Please select your correct option)

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- ○ {}
- ○ Ø
- ● Both {} and Ø

Lec:6,page:16

- ○ Null

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Question No : 5 of 52 Marks: 1 (Budgeted Time 1 Min)

Using tree structure, initial state can be represented by____

Answer (Please select your correct option)

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- ● An arrow

for J.C.Martin tree structure lec:7,page:18

- ○ +
- ● -

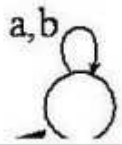
for simple tree structure lec:7,page:17

- ○ *

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Question No : 6 of 52 Marks: 1 (Budgeted Time 1 Min)

Below given FA accepts _____ language.



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Answer (Please select your correct option)

page 17

- Finite
- Infinite
- No
- All

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Question No : 7 of 52 Marks: 1 (Budgeted Time 1 Min)

Even-palindrome is a _____ language

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Answer (Please select your correct option)

- non-regular

conceptual page 70

- regular
- regular but infinite
- regular but finite

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Question No : 8 of 52 Marks: 1 (Budgeted Time 1 Min)

If L1 and L2 are two regular languages, then _____ is/are also regular.

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Answer (Please select your correct option)

- Both $L1 \cup L2$, $L1 \cap L2$

concept

- $L1^c \cap L2$
- $L1 \cup L2$
- $L1 \cap L2$

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Question No : 9 of 52 Marks: 1 (Budgeted Time 1 Min)

Consider the language L of strings, defined over Σ = {a,b}, ending in a

Answer (Please select your correct option)

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- (●) There are finite many classes generated by L, so L is regular

Lec:27,page:76

- ( ) There are infinite many classes generated by L, so L is regular
- ( ) There are finite many classes generated by L, so L is non-regular
- ( ) There are infinite many classes generated by L, so L is non-regular

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Question No : 10 of 52 Marks: 1 (Budgeted Time 1 Min)

Pumping lemma is generally used to prove that:

Answer (Please select your correct option)

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- ( ) A given language is infinite
- (●) A given language is not regular

Conceptual

- ( ) Whether two given regular expressions of a regular language are equivalent or not
- ( ) A given language is finite

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Question No : 11 of 52 Marks: 1 (Budgeted Time 1 Min)

If the FA has N states, then test the words of length less than N. If any one word is accepted by this FA, then it will accept ____ word.

Answer (Please select your correct option)

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- (●) All

Lecture:30,page:85 concept

- ( ) No
- ( ) One
- ( ) Two

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Question No : 12 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following pairs of regular expressions are equivalent?

Answer (Please select your correct option)

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- 1(001)* and (10)*10
- x(xx)* and (x)*x
- x* and x*
- x+ and x*x+

concept

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Question No : 13 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following pairs of regular expressions are not equivalent?

Answer (Please select your correct option)

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- (ab)* and a*b*
- x(xx)* and (xx)*x
- x+ and x*x+
- x(x)* and (x)*x

concept

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Question No : 14 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following statement is correct?

Answer (Please select your correct option)

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- The length of output string is greater than length of input string in moore machine.
- The length of output string is greater than length of input string in mealy machine.
- The length of output string is equal to length of input string in moore machine.
- The length of output string is less than length of input string in mealy machine.

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Question No : 15 of 52 Marks: 1 (Budgeted Time 1 Min)

There exist _____ derivations of a certain word in ambiguous grammar.

Answer (Please select your correct option)

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- ● more than one
- ○ Exactly one
- ○ exactly two
- ○ no

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Question No : 16 of 52 Marks: 1 (Budgeted Time 1 Min)

Which one of the following statements is INCORRECT:

Answer (Please select your correct option)

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- ○ (a+b)*aa(a+b)* generates Regular language.
- ● A language consisting of all strings over ∑={a,b} having equal number of a's and b's is a regular language

Lec:26,page:73

- ○ Every language that can be expressed by FA can also be expressed by RE
- ○ None of these

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Question No : 17 of 52 Marks: 1 (Budgeted Time 1 Min)

We can also represent an FA using different states e.g. Accept state, Reject state, Read State etc.
The _____ state behaves as final state of an FA.

Answer (Please select your correct option)

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- ● Accept

Lec:37,page:105

- ○ Pop
- ○ Push
- ○ Reject

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Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

CYK algorithm was invented by John Cocke and later was published by ____

Answer (Please select your correct option)

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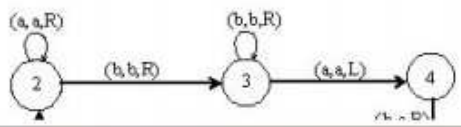
- Talik Khon and Daniel H
- Tandao Kasami and Daniel H
- Tandao Dagger and Daniel H
- Talik Khon

Lec:44,page:135

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Question No : 19 of 52 Marks: 1 (Budgeted Time 1 Min)

Consider the following TM



Answer (Please select your correct option)

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- Above TM accepts the non-CFL {a b c}.
- Above TM accepts the non-CFL $\{a^n b^n a^n\}$.
- Above TM accepts the non-CFL $\{a^n b^{n+2} a^n\}$.
- Above TM accepts the non-CFL $\{a^{n+2} b^n a^n\}$.

Lec:45,page:142

Made by: Waqar Siddhu

Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

Σ= {a,b} then string abbabb has length equal to _____.

Answer (Please select your correct option)

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- Three
- Four
- Five
- Six

No.of letters in string=length

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Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

The language generated by CFG is called _____.

Answer (Please select your correct option)

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- Infinite
- Context Free Language (CFL) Lec:31,page:83
- Regular
- Finite

Made by: Waqar Siddhu

Question No : 22 of 52 Marks: 1 (Budgeted Time 1 Min)

If r1 and r2 are regular expressions then (r1) (r2) is a _____.

Answer (Please select your correct option)

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- RE Lec:3,page:9
- TG
- FA
- GTG

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Question No : 23 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is RE for a language defined over Σ={a,b} having words starting with a

Answer (Please select your correct option)

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- (a + b)*
- aa(a + b)+
- a(a + b)* lec:4,page:12
- a*(a + b)

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Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

The _____ machine helps in building a machine that can perform the addition of binary numbers.

Answer (Please select your correct option)

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- Incrementing

lec:22,page:60

- Complementing
- Decrementing
- Multiplication

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Question No : 25 of 52 Marks: 1 (Budgeted Time 1 Min)

Left hand side of a production in Conext free grammer (CFG) consists of _____.

Answer (Please select your correct option)

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- One terminal
- More than one terminal
- One non-terminal

Lec:31,page:87

- Terminals and non-terminals

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Question No : 26 of 52 Marks: 1 (Budgeted Time 1 Min)

A regular language _____ .

Answer (Please select your correct option)

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- Must be finite
- Must be infinite
- Can be finite or infinite

concept

- Complex Language

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Question No : 27 of 52 Marks: 1 (Budgeted Time 1 Min)

If a language can be expressed by a regular expression, then its complement is _____ .

Answer (Please select your correct option)

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- ● Regular lec:23,page:66
- ○ Non Regular
- ○ Finite
- ○ Infinite

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Question No : 28 of 52 Marks: 1 (Budgeted Time 1 Min)

If a given Context Free grammar (CFG) contains a production of the form S→Λ, then _____

Answer (Please select your correct option)

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- ● The corresponding CFL is not empty. concept
- ○ The corresponding CFL is empty.
- ○ The corresponding CFL is not empty if the language is regular
- ○ The corresponding CFL is empty and non regular.

Made by: Waqar Siddhu

Question No : 29 of 52 Marks: 1 (Budgeted Time 1 Min)

Can a turing machine's head ever be in the same location in two successive steps?

Answer (Please select your correct option)

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- ○ Yes
- ○ Yes but only in finite languages
- ○ No
- ○ Yes but only in infinite langauges

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Question No : 30 of 52 Marks: 1 (Budgeted Time 1 Min)

In _____ there are Finite set of states containing exactly one START state and some (may be none) HALT states that cause execution to terminate, when the HALT states are entered.

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Answer (Please select your correct option)

- TG
- Turing Machine
- GTG
- Finite Automata

Lec 45

Made by: Waqar Siddhu

Question No : 31 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following statement(s) is/are true or false?

1) The turing machine is similar to a finite automaton but with an unlimited and unrestricted memory.
2) A Turing machine is a much more accurate model of a general purpose computer.

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Answer (Please select your correct option)

- Statement 1 is true
- Statement 2 is true
- Both statements (1 & 2) are false
- Statement 2 is false

Both statements true

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Question No : 32 of 52 Marks: 1 (Budgeted Time 1 Min)

We can have more than one parse trees for a given string in case of _____ Context Free Grammar (CFG).

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Answer (Please select your correct option)

- vague
- ambiguous
- hazy
- simple

Lec:33,page:95

Made by: Waqar Siddhu

Question No : 33 of 52 Marks: 1 (Budgeted Time 1 Min)

The language of $a^m b^n$ can be expressed by:

Answer (Please select your correct option)

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- FA
- TG
- GTG
- PDA

concept (non regular language)

Made by: Waqar Siddhu

Question No : 34 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following theorem partitions Σ^* into distinct classes?

Answer (Please select your correct option)

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- Decidability theorem
- Myhill Nerode theorem
- Kleen's theorem
- Emptiness theorem

Lec:27,page:75

Made by: Waqar Siddhu

Question No : 35 of 52 Marks: 1 (Budgeted Time 1 Min)

The languages are supposed to be _____ in case of testing pumping lemma.

Answer (Please select your correct option)

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- context free
- irregular
- both regular and irregular
- regular

Made by: Waqar Siddhu

Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

Consider the following Context Free Grammar (CFG):

S → SXjXY
X → ZZ
Y → 1
Z → 0

Answer (Please select your correct option)

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- 00100
- 0010000
- 001
- 00010

Made by: Waqar Siddhu

Question No : 37 of 52 Marks: 1 (Budgeted Time 1 Min)

A read state can have _____ outgoing edge/ edges.

Answer (Please select your correct option)

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- 1
- 2
- 3
- Any number of

Lec:38,page:112

Made by: Waqar Siddhu

Question No : 38 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following states do not have outgoing edges:

Answer (Please select your correct option)

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- Start
- Read
- Pop
- Accept

Lecture 37

Made by: Waqar Siddhu

Question No : 39 of 52 Marks: 1 (Budgeted Time 1 Min)

Consider the following Context Free Grammar (CFG):

S---->01A

A---->10B

B---->0A|11

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Answer (Please select your correct option)

- ● 011011
- ○ 011011011
- ○ 011001011
- ○ 011001001011

Made by: Waqar Siddhu

Question No : 40 of 52 Marks: 1 (Budgeted Time 1 Min)

Let us consider the following Turing machine (TM):

(y, Y, R)
(f, f, R)
(f, f, L)
(y, y, L)

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Answer (Please select your correct option)

- ○ fffgggg
- ○ fffgggf
- ○ fffggg
- ○ ffffggg

Made by: Waqar Siddhu

Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

What is meant by "Left most Derivation" in context of CFG?

Answer (Please click here to Add Answer)

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Lec:36,page:103

Made by: Waqar Siddhu

Question No : 42 of 52 Marks: 2 (Budgeted Time 4 Min)

Explain the language L of string, defined over ∑= {0,1} , having double 0's or double 1's?

Answer (Please click here to Add Answer)

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(0+1)*(00+11)(0+1)*

Made by: Waqar Siddhu

Question No : 43 of 52 Marks: 2 (Budgeted Time 4 Min)

Consider the language defined by the following regular expression:

b(a + b)*a

What will be the corresponding Context Free Grammar (CFG)?

Answer (Please click here to Add Answer)

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Lec:32,Page:91

Made by: Waqar Siddhu

Question No : 44 of 52 Marks: 2 (Budgeted Time 4 Min)

Consider live productions to be applied during derivation of a word. In your point of view what will be the ultimate result?

Answer (Please click here to Add Answer)

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Lec:43

Made by: Waqar Siddhu

Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

What is meant by the following statement ?

"LANGUAGE IS CLOSED"

Answer (Please click here to Add Answer)

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Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

Write three different parsing techniques.

Answer (Please click here to Add Answer)

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page:136 lec:44

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Question No : 47 of 52 Marks: 3 (Budgeted Time 6 Min)

Illustrate Decidable Problem with the help of an example.

Marks [1+2]

Answer (Please click here to Add Answer)

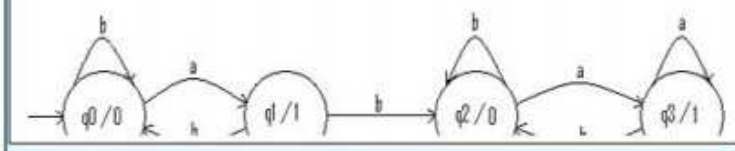
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Lec:29,page80

Made by: Waqar Siddhu

Question No : 48 of 52 Marks: 3 (Budgeted Time 6 Min)

Consider the following Moore machine:



Answer (Please click here to Add Answer)

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Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

Consider the language L which is EVEN-EVEN, defined over Σ = {a,b}. In how many classes does L may partition Σ*. Explain briefly.

Answer (Please click here to Add Answer)

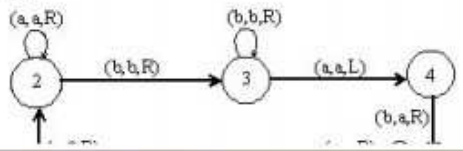
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Lec:28,Page:77

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Question No : 50 of 52 Marks: 5 (Budgeted Time 10 Min)

Consider the following TM



Answer (Please click here to Add Answer)

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Question No : 51 of 52 **Marks: 5 (Budgeted Time 10 Min)**

Derive Chomsky Normal Form (CNF) corresponding to the Context Free Grammar (CFG) given below:

$A \rightarrow BAB \mid B \mid \Lambda$

$B \rightarrow 00 \mid \Lambda$

Answer (Please click here to Add Answer)

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Question No : 52 of 52 **Marks: 5 (Budgeted Time 10 Min)**

Suppose a Maze of 20 boxes; 1 box for initial and 1 for final state. 6 boxes are for non-moveable states while the rest can be used for transitions. Draw a Nondeterministic Finite Automaton (NFA) for the given Maze.

Marks [2.5+2.5]

Answer (Please click here to Add Answer)

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بِسْمِ اللّٰهِ الرَّحْمٰنِ الرَّحِيْمِ

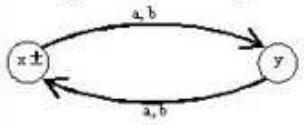
In the Name of Allāh, the Most Gracious, the Most Merciful

Paper Pattern

MCQS 40 each 1 mark
Short 4 each 2 marks
Short 4 each 3 marks
long 4 each 5 marks

Question No : 2 of 52 Marks: 1 (Budgeted Time 1 Min)

Below given FA can be represented by



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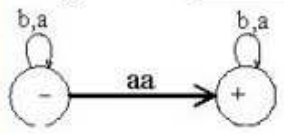
Answer (Please select your correct option)

- ((a+ b) (a + b))*
- (a + b)(a + b)*
- (a + b)(a + b)
- (a + b)*(a + b)*

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Question No : 3 of 52 Marks: 1 (Budgeted Time 1 Min)

Below given TG has _____ RE



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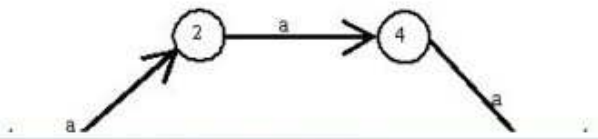
Answer (Please select your correct option)

- (b+a) (aa) (b+a)*
- (a+b) (aa) (a+b)*
- (a+b)* (aa) (a+b)*
- (a+b)* (aa) (a+b)

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Question No : 4 of 52 Marks: 1 (Budgeted Time 1 Min)

Below given TG's are _____.



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Answer (Please select your correct option)

- non-symmetric
- Equivalent
- Non-equivalent
- TG's are not valid

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Question No : 5 of 52 Marks: 1 (Budgeted Time 1 Min)

"One GTG can generate _____ language".

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Answer (Please select your correct option)

- more than one
- only one
- only two
- only three

Made by: Waqar Siddhu

Question No : 6 of 52 Marks: 1 (Budgeted Time 1 Min)

Even-palindrome is a _____ language

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Answer (Please select your correct option)

- non-regular
- regular
- regular but infinite
- regular but finite

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Question No : 7 of 52 Marks: 1 (Budgeted Time 1 Min)

If L1 and L2 are two regular languages, then _____ is/are also regular.

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Answer (Please select your correct option)

- Both L1 ∪ L2 , L1 ∩ L2
- $L1^c \cap L2$
- L1 ∪ L2
- L1 ∩ L2

Made by: Waqar Siddhu

Question No : 8 of 52 Marks: 1 (Budgeted Time 1 Min)

If the FA has N states, then test the words of length less than N. If any one word is accepted by this FA, then it will accept ____ word.

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Answer (Please select your correct option)

- All
- No
- One
- Two

Made by: Waqar Siddhu

Question No : 9 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following pairs of regular expressions are equivalent?

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Answer (Please select your correct option)

- 1(001)* and (10)*10
- x(xx)* and (x)*x
- x^+ and x^*
- x^+ and x^*x^+

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Question No : 10 of 52 Marks: 1 (Budgeted Time 1 Min)

An alphabet of Σ is valid if

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Answer (Please select your correct option)

- No letter of Σ appears in middle of any other letter
- No letter of Σ appears at end of any other letter
- No letter of Σ appears at start of any other letter
- No letter of Σ appears at end or middle of any other letter

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Question No : 11 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following pairs of regular expressions are not equivalent?

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Answer (Please select your correct option)

- (ab)* and a*b*
- x(xx)* and (xx)*x
- x^+ and x^*x^+
- x(x)* and (x)*x

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Question No : 12 of 52 Marks: 1 (Budgeted Time 1 Min)

There exist _____ derivations of a certain word in ambiguous grammar.

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Answer (Please select your correct option)

- more than one
- Exactly one
- exactly two
- no

Made by: Waqar Siddhu

Question No : 13 of 52 Marks: 1 (Budgeted Time 1 Min)

Regular languages are closed under _____

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Answer (Please select your correct option)

- Union and Concatenation
- Union and Division
- Union and Exponent
- Union and Factorial

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Question No : 14 of 52 Marks: 1 (Budgeted Time 1 Min)

Which one of the following statements is INCORRECT.

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Answer (Please select your correct option)

- A language consisting of all strings over ∑={a,b} having equal number of a's and b's is a regular language
- Every language that can be expressed by FA can also be expressed by RE
- None of these
- (a+b)*aa(a+b)* generates Regular language.

Made by: Waqar Siddhu

Question No : 15 of 52 Marks: 1 (Budgeted Time 1 Min)

TM is more powerful than FSM because

not sure

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Answer (Please select your correct option)

- The tape movement is confined to one direction
- It has no finite state control
- It has the capability to remember arbitrary long sequences of input symbols
- The tape movement is confined to more than one directions

Made by: Waqar Siddhu

Question No : 16 of 52 Marks: 1 (Budgeted Time 1 Min)

CYK algorithm was invented by John Cocke and later was published by ____

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Answer (Please select your correct option)

- Talik Khon and Daniel H
- Tandao Kasami and Daniel H
- Tandao Dagger and Daniel H
- Talik Khon

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Question No : 17 of 52 Marks: 1 (Budgeted Time 1 Min)

The process of finding the derivation of word generated by particular grammar is called_____

pg 136

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Answer (Please select your correct option)

- Processing
- Parsing
- Programming
- Planing

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Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

There are different parsing techniques, including _____

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Answer (Please select your correct option)

- Top down parsing
- Bottom up parsing
- Parsing technique for particular grammar of arithmetic expression
- All of these

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Question No : 19 of 52 Marks: 1 (Budgeted Time 1 Min)

Σ={a,Aa,Abb} , then string aAaAbbAa has length equal to_____.

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Answer (Please select your correct option)

- Three
- Four
- One
- Two

Made by: Waqar Siddhu

Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

The language generated by CFG is called _____.

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Answer (Please select your correct option)

- Infinite
- Context Free Language (CFL)
- Regular
- Finite

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Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

In _____ there must be transition for all letters.

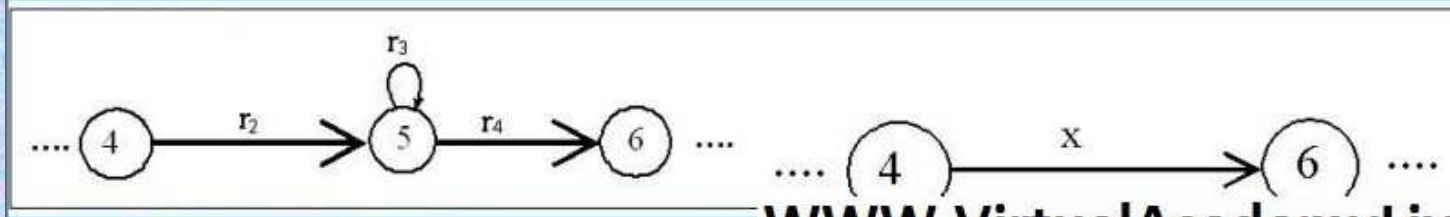
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Answer (Please select your correct option)

- NFA
- FA
- TG
- GTG

Made by: Waqar Siddhu

Question No : 22 of 52 Marks: 1 (Budgeted Time 1 Min)



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Answer (Please select your correct option)

- r1*r2 r3*
- (r1r2r3)*
- r1*r2*r3*
- None of the given

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Question No : 23 of 52 Marks: 1 (Budgeted Time 1 Min)

A regular language _____ .

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Answer (Please select your correct option)

- Must be finite
- Must be infinite
- Can be finite or infinite
- Complex Language

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Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

If a language can be expressed by a regular expression, then its complement is _____ .

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Answer (Please select your correct option)

- Regular
- Non Regular
- Finite
- Infinite

Made by: Waqar Siddhu

Question No : 25 of 52 Marks: 1 (Budgeted Time 1 Min)

For every Context Free Grammar (CFG), we can make the corresponding _____.

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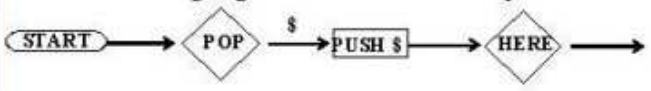
Answer (Please select your correct option)

- FA
- TG
- PDA
- Regular Grammar 

Made by: Waqar Siddhu

Question No : 26 of 52 Marks: 1 (Budgeted Time 1 Min)

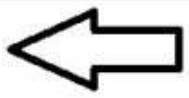
Look at the following diagram and select the correct option:



pg 120

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Answer (Please select your correct option)

- The PDA must not begin with the sequence
- The PDA may begin with the sequence
- The PDA must begin with the sequence 
- The PDA may not begin with the sequence

Made by: Waqar Siddhu

Question No : 27 of 52 Marks: 1 (Budgeted Time 1 Min)

Consider the following Context Free Grammar (CFG) and select the correct choice

$S \rightarrow AB$

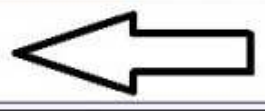
$A \rightarrow BSB \mid a \mid b$

$B \rightarrow CC$

$C \rightarrow SS \mid b \mid bb$

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Answer (Please select your correct option)

- abb is a word in the corresponding CFL. 
- abb is not the word of corresponding CFL.
- Any word can be accepted from the corresponding CFL.
- b is a word in the corresponding CFL.

Made by: Waqar Siddhu

Question No : 28 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following statement(s) is/are true or false?

1) The turing machine is similar to a finite automaton but with an unlimited and unrestricted memory.
2) A Turing machine is a much more accurate model of a general purpose computer.

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Answer (Please select your correct option)

- Statement 1 is true
- Statement 2 is true
- Both statements (1 & 2) are false
- Statement 2 is false

Made by: Waqar Siddhu

Question No : 29 of 52 Marks: 1 (Budgeted Time 1 Min)

Language of strings having even number of a's is considered to be:

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Answer (Please select your correct option)

- Regular
- Irrelgular
- Irregular and infinite
- Irregular and finite

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Question No : 30 of 52 Marks: 1 (Budgeted Time 1 Min)

If we have two transiton graphs then their union will be expressed by:

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Answer (Please select your correct option)

- taking a common start state and joining them by two null transitions
- just connecting both start states by null transitions
- connecting final state of first TG to the initial state of second TG
- connecting the final state of first TG to the final state of second TG

Made by: Waqar Siddhu

Question No : 31 of 52 Marks: 1 (Budgeted Time 1 Min)

A language of strings of prime length is considered to be:

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Answer (Please select your correct option)

- regular but infinite
- regular
- irregular
- regular but finite

Made by: Waqar Siddhu

Question No : 32 of 52 Marks: 1 (Budgeted Time 1 Min)

_____ and _____ are removed in order to make a CFG in Chomsky Normal Form (CNF).

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Answer (Please select your correct option)

- Null, nullable productions
- Nullable, unit productions
- Null, unit productions
- Strings of length 0, null

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Question No : 33 of 52 Marks: 1 (Budgeted Time 1 Min)

The languages are supposed to be _____ in case of testing pumping lemma.

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Answer (Please select your correct option)

- context free
- irregular
- both regular and irregular
- regular

Made by: Waqar Siddhu

Question No : 34 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following refers to an effectively solvable problem having answer in "Yes" or "No"?

pg 80

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Answer (Please select your correct option)

- Decision procedure
- Logical problem
- Analytical problem
- Binary procedure

Made by: Waqar Siddhu

Question No : 35 of 52 Marks: 1 (Budgeted Time 1 Min)

Consider the following Context Free Grammar (CFG)

S ---> SX|XY
X ---> ZZ
Y ---> 0
Z ---> 1

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Answer (Please select your correct option)

- 110
- 11011
- 1101111
- 11101

Made by: Waqar Siddhu

Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

Consider the following Context Free Grammar (CFG)

S → SX|XY
X → ZZ
Y → 1
Z → 0

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Answer (Please select your correct option)

- 001
- 00010
- 0010000
- 00100

Made by: Waqar Siddhu

Question No : 37 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is a regular Context Free Grammar:

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Answer (Please select your correct option)

- S→abS | baS | ^
- S→aSb | baS |^
- S→abS | bSa | ^
- S→aSb | Sa | ^

Made by: Waqar Siddhu

Question No : 38 of 52 Marks: 1 (Budgeted Time 1 Min)

Concatenation of r1 and r2 will be shown as:

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Answer (Please select your correct option)

- r1+r2
- r2+r1
- r2r1
- r1r2

Made by: Waqar Siddhu

Question No : 39 of 52 Marks: 1 (Budgeted Time 1 Min)

A nonterminal is said to be self-embedded, if in a given derivation of a word, it ever occurs as a tree _____ of itself.

pg 126 bk

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Answer (Please select your correct option)

- ascendant
- descendant
- top
- bottom

Made by: Waqar Siddhu

Question No : 40 of 52 Marks: 1 (Budgeted Time 1 Min)

_____ is always Deterministic

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Answer (Please select your correct option)

- Finite Automaton
- Transition Graph
- Generalize Transition Graph
- Non-deterministic finite automaton

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Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

Write alternative form of this production.
$S \rightarrow aS, S \rightarrow bS, S \rightarrow \Lambda$

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Answer (Please click here to Add Answer)

Made by: Waqar Siddhu

Question No : 42 of 52 Marks: 2 (Budgeted Time 4 Min)

What are the uses of Push Down Automata (PDA) in computing?

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Answer (Please click here to Add Answer)

The PDA is used in theories about what can be computed by machines. PDA described with a formal grammar, it can be used in parser design. The deterministic pushdown automaton can handle all deterministic context-free languages while the nondeterministic version can handle all context-free languages.

Made by: Waqar Siddhu

Question No : 43 of 52 Marks: 2 (Budgeted Time 4 Min)

What is meant by Row language?

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Answer (Please click here to Add Answer)

row language whose alphabet is $\Sigma = \{Row_1, Row_2, \ldots, Row_7\}$ *i.e.* the alphabet consists of the letters which are the names of the rows in the summary table.

that the Row language is the language whose alphabet $\Sigma = \{Row_1, Row_2, \ldots, Row_7\}$,

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Question No : 44 of 52 Marks: 2 (Budgeted Time 4 Min)

Write down regular expression for a language L of strings defined over $\Sigma = (a, b)$ that neither ends in "a" nor with "b".

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Answer (Please click here to Add Answer)

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Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

What are the steps of Recursive Definition of Languages?

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Answer (Plea

Recursive definition of languages

The following three steps are used in recursive definition

Some basic words are specified in the language.

Rules for constructing more words are defined in the language.

No strings except those constructed in above, are allowed to be in the language.

Examples

Defining language of INTEGER

Step 1: 1 is in **INTEGER**.

Step 2: If x is in **INTEGER** then x+1 and x-1 are also in **INTEGER**.

Step 3: No strings except those constructed in above, are allowed to be in **INTEGER**.

Made by: Waqar Siddhu

Question No : 46 of 52 **Marks: 3 (Budgeted Time 6 Min)**

What does the following arbitrary summary table show?

FROM Where	TO Where	READ What	POP What	PUSH What	ROW Number
$READ_9$	$READ_3$	b	b	abb	11

Answer (Please click here to Add Answer)

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Consider the following row from an arbitrary summary ta

FROM Where	TO Where	READ What	POP What	PUSH What	ROW Number
$READ_9$	$READ_3$	b	b	abb	11

which shows that Row_{11} is not Net style sentence because the trip from $READ_9$ to $READ_3$ does not pop one letter form the STACK, while it adds two letters to the STACK. However Row_{11} can be concatenated with some other Net style sentences *e.g.* $Row_{11}Net(READ_3, READ_7, a)Net(READ_7, READ_1, b)Net(READ_1, READ_8, b)$
Which gives the nonterminal
$Net(READ_9, READ_8, b)$, now the whole process can be written as
$Net(READ_9, READ_8, b) \rightarrow Row_{11}Net(READ_3, READ_7, a)\ Net(READ_7, READ_1, b)Net(READ_1, READ_8, b)$
Which will be a production in the CFG of the corresponding **row language**.

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Question No : 47 of 52 **Marks: 3 (Budgeted Time 6 Min)**

Write down Context Free Grammar (CFG) for the language of strings expressed by the following regular expression:

$(a + b)^*(a + b)^*$

Answer (Please click here to Add Answer)

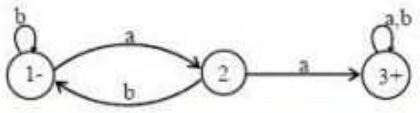
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S->a|b|Xa|Xb, X->aX|bX|^

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Question No : 48 of 52 **Marks: 3 (Budgeted Time 6 Min)**

Let us consider the following Finite Automaton (FA):



Answer (Please click here to Add Answer)

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Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

Write the CFG for EVEN-EVEN language?

Answer (Please click here to Add Ansv

Example

$\Sigma = \{a,b\}$

productions:

$S \rightarrow SS$

$S \rightarrow XS$

$S \rightarrow \Lambda$

$S \rightarrow YSY$

$X \rightarrow aa$

$X \rightarrow bb$

$Y \rightarrow ab$

$Y \rightarrow ba$

This grammar generates EVEN-EVEN language.

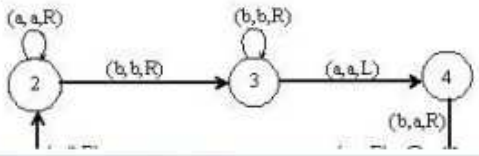
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Question No : 50 of 52 Marks: 5 (Budgeted Time 10 Min)

Consider the following TM



Answer (Please click here to Add Answer)

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Question No : 51 of 52 Marks: 5 (Budgeted Time 10 Min)

Using suitable examples, differentiate between Kleene star closure and plus operation?

Answer (Pleas

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Kleene Star Closure

Given Σ, then the Kleene Star Closure of the alphabet over Σ, including Λ.

It is to be noted that Kleene Star Closure can be defined over any set of strings.

Examples

If $\Sigma = \{x\}$
Then $\Sigma^* = \{\Lambda, x, xx, xxx, xxxx, ...\}$
If $\Sigma = \{0,1\}$
Then $\Sigma^* = \{\Lambda, 0, 1, 00, 01, 10, 11, ...\}$
If $\Sigma = \{aaB, c\}$
Then $\Sigma^* = \{\Lambda, aaB, c, aaBaaB, aaBc, caaB, cc, ...\}$

Note

Languages generated by Kleene Star Closure of set of strings, are infinite languages. (By infinite language, it is supposed that the language contains infinite many words, each of finite length).

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PLUS Operation ($^+$)

Plus Operation is same as Kleene Star Closure except that it does not generate Λ (null string), automatically.

Example
If $\Sigma = \{0,1\}$
Then $\Sigma^+ = \{0, 1, 00, 01, 10, 11, ...\}$
If $\Sigma = \{aab, c\}$
Then $\Sigma^+ = \{aab, c, aabaab, aabc, caab, cc, ...\}$

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Question No : 52 of 52 Marks: 5 (Budgeted Time 10 Min)

a. Construct Context Free Grammar (CFG) for the language defined by the regular expression:

$(ab + ba)^*(a + b)^*$

b. Remove null productions from the CFG constructed in part i.

Answer (Please click here to Add Answer)

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Paper Pattern

MCQS 40 each 1 mark
Short 4 each 2 marks
Short 4 each 3 marks
long 4 each 5 marks

Question No : 1 of 52 Marks: 1 (Budgeted Time 1 Min)

Which statement is true?

pg 4

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Answer (Please select your correct option)

- All words are strings
- All strings are words
- Both are always same
- All letter are string

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Question No : 2 of 52 Marks: 1 (Budgeted Time 1 Min)

If L1 and L2 are expressed by r1 and r2, respectively then the language expressed by (r1)(r2) will be _____

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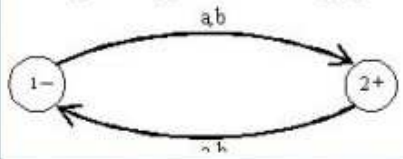
Answer (Please select your correct option)

- Regular
- Ir-regular
- Undermined
- Such language does not exist

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Question No : 3 of 52 Marks: 1 (Budgeted Time 1 Min)

Below given FA generates the language having strings of _____



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Answer (Please select your correct option)

- ODD length
- EVEN length
- Equal number of a's and b's
- Odd number of a's and b's

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Question No : 4 of 52 Marks: 1 (Budgeted Time 1 Min)

If an FA accepts no string, then language is denoted by _____

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Answer (Please select your correct option)

- {}
- Ø
- Both { } and Ø
- Null

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Question No : 5 of 52 Marks: 1 (Budgeted Time 1 Min)

Using tree structure, initial state can be represented by____

pg 16 bk

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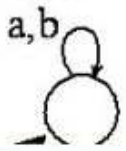
Answer (Please select your correct option)

- An arrow
- +
- -
- *

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Question No : 6 of 52 Marks: 1 (Budgeted Time 1 Min)

Below given FA accepts _____ language.



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Answer (Please select your correct option)

- Finite
- Infinite
- No
- All

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Question No : 7 of 52 Marks: 1 (Budgeted Time 1 Min)

Even-palindrome is a _____ language

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Answer (Please select your correct option)

- non-regular ⇦
- regular
- regular but infinite
- regular but finite

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Question No : 8 of 52 Marks: 1 (Budgeted Time 1 Min)

If L1 and L2 are two regular languages, then _____ is/are also regular.

pg 63 & 66 of bk

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Answer (Please select your correct option)

- Both $L1 \cup L2$, $L1 \cap L2$ ⇦
- $L1' \cap L2$
- $L1 \cup L2$
- $L1 \cap L2$

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Question No : 9 of 52 Marks: 1 (Budgeted Time 1 Min)

Consider the language L of strings, defined over Σ = {a,b}, ending in a

pg 74 bk

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Answer (Please select your correct option)

- There are finite many classes generated by L, so L is regular
- There are infinite many classes generated by L, so L is regular
- There are finite many classes generated by L, so L is non-regular
- There are infinite many classes generated by L, so L is non-regular

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Question No : 10 of 52 Marks: 1 (Budgeted Time 1 Min)

Pumping lemma is generally used to prove that:

pg 72

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Answer (Please select your correct option)

- A given language is infinite
- A given language is not regular
- Whether two given regular expressions of a regular language are equivalent or not
- A given language is finite

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Question No : 11 of 52 Marks: 1 (Budgeted Time 1 Min)

If the FA has N states, then test the words of length less than N. If any one word is accepted by this FA, then it will accept ____ word.

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Answer (Please select your correct option)

- All

not sure

- No
- One
- Two

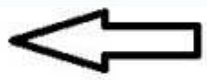
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Question No : 12 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following pairs of regular expressions are equivalent?

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Answer (Please select your correct option)

- 1(001)* and (10)*10
- x(xx)* and (x)*x
- x^+ and x* 
- x^+ and x^*x^+

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Question No : 13 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following pairs of regular expressions are not equivalent?

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Answer (Please select your correct option)

- (ab)* and a*b* 
- x(xx)* and (xx)*x
- x^+ and x^*x^+
- x(x)* and (x)*x

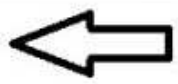
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Question No : 14 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following statement is correct?

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Answer (Please select your correct option)

- The length of output string is greater than length of input string in moore machine. 
- The length of output string is greater than length of input string in mealy machine.
- The length of output string is equal to length of input string in moore machine.
- The length of output string is less than length of input string in mealy machine.

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Question No : 15 of 52 Marks: 1 (Budgeted Time 1 Min)

There exist _____ derivations of a certain word in ambiguous grammar.

pg 95

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Answer (Please select your correct option)

- more than one
- Exactly one
- exactly two
- no

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Question No : 16 of 52 Marks: 1 (Budgeted Time 1 Min)

Which one of the following statements is INCORRECT:

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Answer (Please select your correct option)

- (a+b)*aa(a+b)* generates Regular language.
- A language consisting of all strings over ∑={a,b} having equal number of a's and b's is a regular language
- Every language that can be expressed by FA can also be expressed by RE
- None of these

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Question No : 17 of 52 Marks: 1 (Budgeted Time 1 Min)

We can also represent an FA using different states e.g. Accept state, Reject state, Read State etc.
The _____ state behaves as final state of an FA.

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Answer (Please select your correct option)

- Accept
- Pop
- Push
- Reject

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Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

CYK algorithm was invented by John Cocke and later was published by ____

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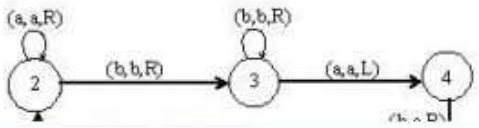
Answer (Please select your correct option)

- Talik Khon and Daniel H
- Tandao Kasami and Daniel H
- Tandao Dagger and Daniel H
- Talik Khon

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Question No : 19 of 52 Marks: 1 (Budgeted Time 1 Min)

Consider the following TM



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Answer (Please select your correct option)

- Above TM accepts the non-CFL {a b c}.
- Above TM accepts the non-CFL $\{a^n b^n a^n\}$.
- Above TM accepts the non-CFL $\{a^n b^{n+2} a^n\}$.
- Above TM accepts the non-CFL $\{a^{n+2} b^n a^n\}$.

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Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

Σ= (a,b) then string abbabb has length equal to _____.

pg 4

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Answer (Please select your correct option)

- Three
- Four
- Five
- Six

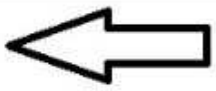
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Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

The language generated by CFG is called _____.

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Answer (Please select your correct option)

- Infinite
- Context Free Language (CFL) 
- Regular
- Finite

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Question No : 22 of 52 Marks: 1 (Budgeted Time 1 Min)

If r1 and r2 are regular expressions then (r1) (r2) is a _____ .

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Answer (Please select your correct option)

- RE 
- TG
- FA
- GTG

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Question No : 23 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is RE for a language defined over Σ=(a,b) having words starting with a:

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Answer (Please select your correct option)

- (a + b)*
- aa(a + b)+
- a(a + b)* 
- a*(a + b)

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Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

The _____ machine helps in building a machine that can perform the addition of binary numbers.

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Answer (Please select your correct option)

- Incrementing
- Complementing
- Decrementing
- Multiplication

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Question No : 25 of 52 Marks: 1 (Budgeted Time 1 Min)

Left hand side of a production in Conext free grammer (CFG) consists of _____.

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Answer (Please select your correct option)

- One terminal
- More than one terminal
- One non-terminal
- Terminals and non-terminals

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Question No : 26 of 52 Marks: 1 (Budgeted Time 1 Min)

A regular language _____.

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Answer (Please select your correct option)

- Must be finite
- Must be infinite
- Can be finite or infinite
- Complex Language

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Question No : 27 of 52 Marks: 1 (Budgeted Time 1 Min)

If a language can be expressed by a regular expression, then its complement is _____ .

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Answer (Please select your correct option)

- Regular
- Non Regular
- Finite
- Infinite

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Question No : 28 of 52 Marks: 1 (Budgeted Time 1 Min)

If a given Context Free grammar (CFG) contains a production of the form $S \rightarrow \Lambda$, then _____

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Answer (Please select your correct option)

- The corresponding CFL is not empty.
- The corresponding CFL is empty.
- The corresponding CFL is not empty if the language is regular
- The corresponding CFL is empty and non regular.

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Question No : 29 of 52 Marks: 1 (Budgeted Time 1 Min)

Can a turing machine's head ever be in the same location in two successive steps?

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Answer (Please select your correct option)

- Yes
- Yes but only in finite languages
- No
- Yes but only in infinite langauges

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Question No : 30 of 52 Marks: 1 (Budgeted Time 1 Min)

In _____ there are Finite set of states containing exactly one START state and some (may be none) HALT states that cause execution to terminate, when the HALT states are entered.

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Answer (Please select your correct option)

- TG
- Turing Machine
- GTG
- Finite Automata

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Question No : 31 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following statement(s) is/are true or false?

1) The turing machine is similar to a finite automaton but with an unlimited and unrestricted memory.
2) A Turing machine is a much more accurate model of a general purpose computer.

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Answer (Please select your correct option)

- Statement 1 is true
- Statement 2 is true
- Both statements (1 & 2) are false
- Statement 2 is false

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Question No : 32 of 52 Marks: 1 (Budgeted Time 1 Min)

We can have more than one parse trees for a given string in case of _____ Context Free Grammar (CFG).

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Answer (Please select your correct option)

- vague
- ambiguous
- hazy
- simple

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Question No : 33 of 52 Marks: 1 (Budgeted Time 1 Min)

The language of $a^n b^n$ can be expressed by:

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Answer (Please select your correct option)

- FA
- TG
- GTG
- PDA



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Question No : 34 of 52 Marks: 1 (Budgeted Time 1 Min)

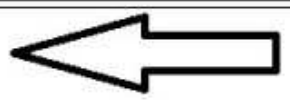
Which of the following theorem partitions Σ^* into distinct classes?

pg 73 bk

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Answer (Please select your correct option)

- Decidability theorem
- Myhill Nerode theorem



- Kleen's theorem
- Emptiness theorem

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Question No : 35 of 52 Marks: 1 (Budgeted Time 1 Min)

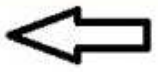
The languages are supposed to be _____ in case of testing pumping lemma.

to prove puming lemma theorem language is suposed to be regular at first and thn proof it non regular by pumping lemma

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Answer (Please select your correct option)

- context free
- irregular
- both regular and irregular
- regular



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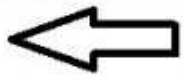
Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

Consider the following Context Free Grammar (CFG)

S → SX|XY
X → ZZ
Y → 1
Z → 0

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Answer (Please select your correct option)

- 00100 
- 0010000
- 001
- 00010

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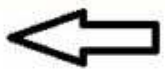
Question No : 37 of 52 Marks: 1 (Budgeted Time 1 Min)

A read state can have _____ outgoing edge/ edges.

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Answer (Please select your correct option)

- 1
- 2
- 3
- Any number of 

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Question No : 38 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following states do not have outgoing edges:

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Answer (Please select your correct option)

- Start
- Read
- Pop
- Accept 

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Question No : 39 of 52 Marks: 1 (Budgeted Time 1 Min)

Consider the following Context Free Grammar (CFG)

S---->01A

A---->10B

B---->0A|11

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Answer (Please select your correct option)

- 011011
- 011011011
- 011001011
- 011001001011

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Question No : 40 of 52 Marks: 1 (Budgeted Time 1 Min)

Let us consider the following Turing machine (™):

(y, Y, R)
(f, f, R)
(f, f, L)
(v, v, L)

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Answer (Please select your correct option)

- fffgggg
- fffggggf
- fffggg
- ffffggg

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Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

What is meant by "Left most Derivation" in context of CFG?

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Answer (Please click here to Add Answer)

Left most derivation

Definition

The derivation of a word w, generated by a CFG, such that at each step, a production is applied to the left most nonterminal in the working string, is said to be ***left most derivation***.

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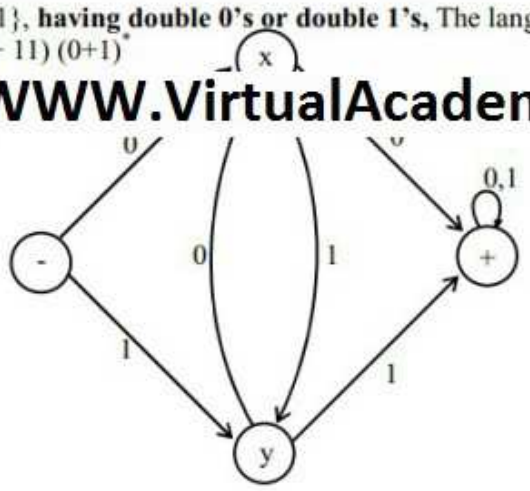
Question No : 42 of 52 — **Marks: 2 (Budgeted Time 4 Min)**

Explain the language L of string, defined over ∑= {0,1}, having double 0's or double 1's?

Consider the language L of strings, defined over Σ={0, 1}, **having double 0's or double 1's,** The language L may be expressed by the regular expression $(0+1)^*(00 + 11)(0+1)^*$
This language may be accepted by the following FA

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Answer (F



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Question No : 43 of 52 — **Marks: 2 (Budgeted Time 4 Min)**

Consider the language defined by the following regular expression:

b(a + b)*a

What will be the corresponding Context Free Grammar (CFG)?

Answer (Please click here to Add Answer)

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Consider the following CFG
S → aXb|bXa
X → aX|bX|Λ
The above CFG generates the language of strings, defined over ∑={a,b}, **beginning and ending in different letters.**

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Question No : 44 of 52 — **Marks: 2 (Budgeted Time 4 Min)**

Consider live productions to be applied during derivation of a word. In your point of view what will be the ultimate result?

Answer (Please click here to Add Answer)

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It may be noted that every time a live production is applied during the derivation of a word it increases the number of nonterminals by one.

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Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

What is meant by the following statement ?

"LANGUAGE IS CLOSED"

Answer (Please click here to Add Answer)

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$\Sigma = \{a,b\}$
productions:

$S \rightarrow SS$
$S \rightarrow XS$
$S \rightarrow \Lambda$

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Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

Write three different parsing techniques.

Answer (Please click here to Add Answer)

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Parsing of word

Definition

The process of finding the derivation of word generated by particular grammar is called **parsing**.
There are different parsing techniques, containing the following three
Top down parsing.
Bottom up parsing.
Parsing technique for particular grammar of arithmetic expression.

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Question No : 47 of 52 Marks: 3 (Budgeted Time 6 Min)

Illustrate Decidable Problem with the help of an example.

Marks [1+2]

Answer (Please click here to Add Answer)

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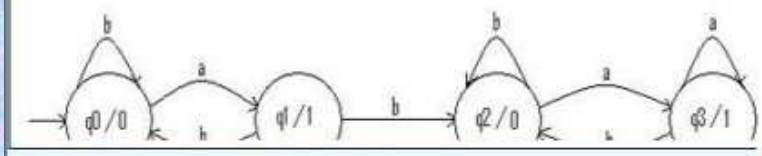
Decidable problem

A problem that has decision procedure is called decidable problem *e.g.* the following problems
The two regular expressions define the same language.
The two FAs are equivalent.

Made by: Waqar Siddhu

Question No : 48 of 52 — **Marks: 3 (Budgeted Time 6 Min)**

Consider the following Moore machine:



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Answer (Please click here to Add Answer)

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Question No : 49 of 52 — **Marks: 5 (Budgeted Time 10 Min)**

Consider the language L which is EVEN-EVEN, defined over $\Sigma = \{a,b\}$. In how many classes does L may partition Σ^*. Explain briefly.

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Answer (Please click here to Add Answer)

Example

Consider the language L which is EVEN-EVEN, defined over $\Sigma = \{a,b\}$. It can be observed that L partitions Σ^* into the following four classes

C_1 = set of all strings with even number of a's and odd number of b's.

C_2 = set of all strings with odd number of a's and odd number of b's.

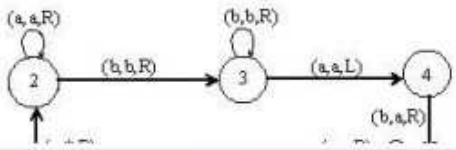
C_3 = set of all strings with odd number of a's and even number of b's.

C_4 = set of all strings with even number of a's and even number of b's.

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Question No : 50 of 52 — **Marks: 5 (Budgeted Time 10 Min)**

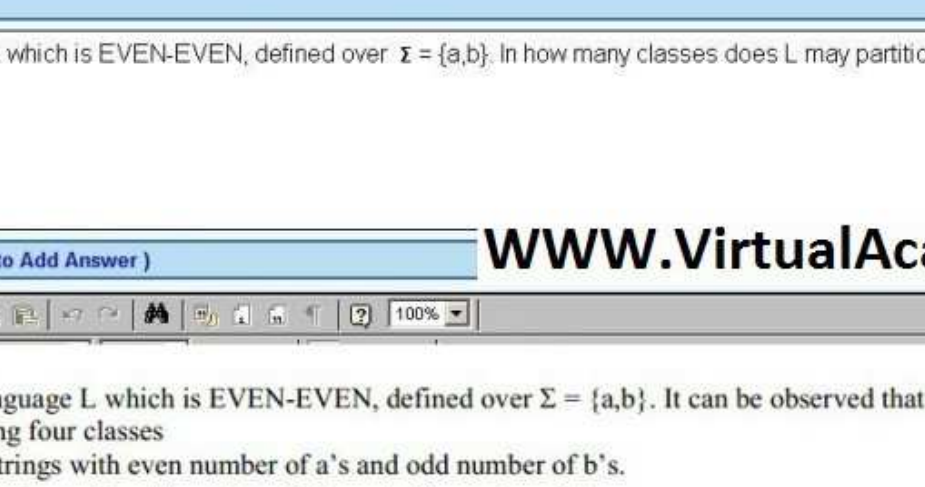
Consider the following TM



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Answer (Please click here to Add Answer)

Consider the following TM



The string aaabbbaaa can be observed to be accepted by the above TM. It can also be observed that the above TM accepts the non-CFL $\{a^n b^n a^n\}$.

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Question No : 51 of 52 — **Marks: 5 (Budgeted Time 10 Min)**

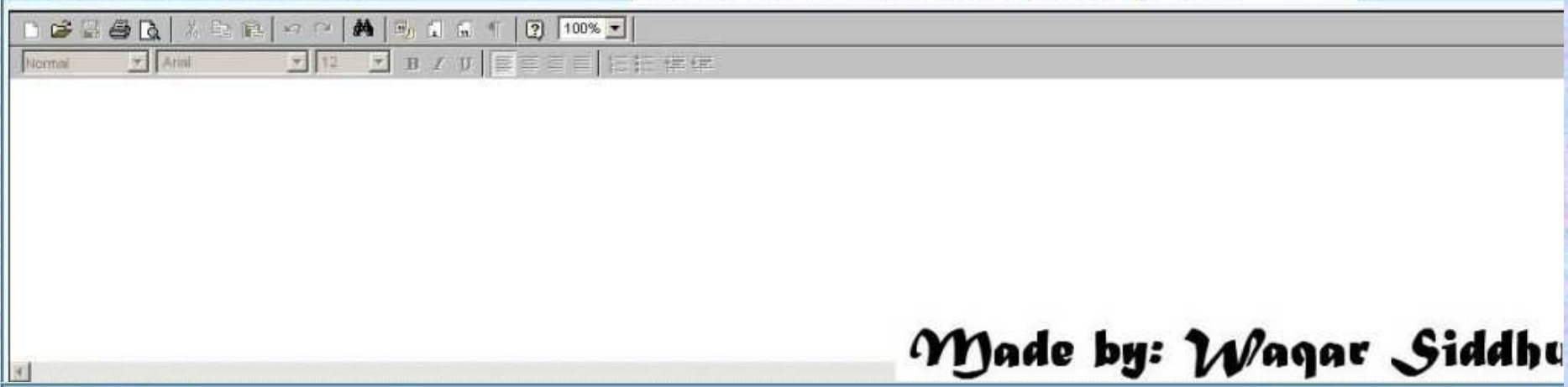
Derive Chomsky Normal Form (CNF) corresponding to the Context Free Grammar (CFG) given below:

$A \rightarrow BAB \mid B \mid \Lambda$

$B \rightarrow 00 \mid \Lambda$

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Answer (Please click here to Add Answer)



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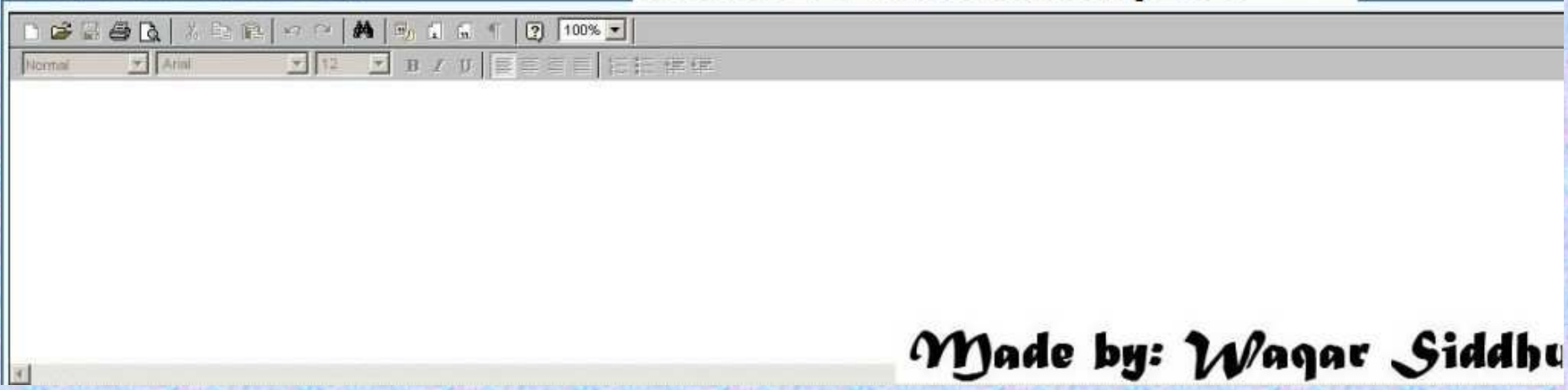
Question No : 52 of 52 — **Marks: 5 (Budgeted Time 10 Min)**

Suppose a Maze of 20 boxes; 1 box for initial and 1 for final state. 6 boxes are for non-moveable states while the rest can be used for transitions. Draw a Nondeterministic Finite Automaton (NFA) fo the given Maze.

Marks [2.5+2.5]

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Answer (Please click here to Add Answer)



Made by: Waqar Siddhu

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بِسْمِ اللّٰهِ الرَّحْمٰنِ الرَّحِيْمِ

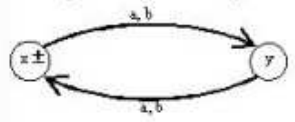
In the Name of Allāh, the Most Gracious, the Most Merciful

Paper Pattern

MCQS 40 each 1 mark
Short 4 each 2 marks
Short 4 each 3 marks
long 4 each 5 marks

Question No : 2 of 52 Marks: 1 (Budgeted Time 1 Min)

Below given FA can be represented by



Answer (Please select your correct option)

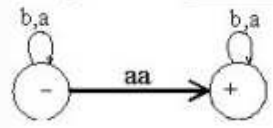
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- ((a+ b) (a + b))* lec:5 ,page:13
- (a + b)(a + b)*
- (a + b)(a + b)
- (a + b)*(a + b)*

Made by: Waqar Siddhu

Question No : 3 of 52 Marks: 1 (Budgeted Time 1 Min)

Below given TG has _____ RE



Answer (Please select your correct option)

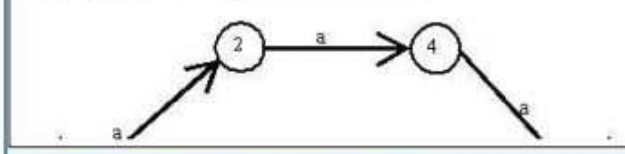
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- (b+a) (aa) (b+a)*
- (a+b) (aa) (a+b)*
- (a+b)* (aa) (a+b)* lec:8,page:20
- (a+b)* (aa) (a+b)

Made by: Waqar Siddhu

Question No : 4 of 52 Marks: 1 (Budgeted Time 1 Min)

Below given TG's are _____.



Answer (Please select your correct option)

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- non-symmetric
- Equivalent
- Non-equivalent
- TG's are not valid

page 21

Made by: Waqar Siddhu

Question No : 5 of 52 Marks: 1 (Budgeted Time 1 Min)

"One GTG can generate _____ language".

Answer (Please select your correct option)

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- more than one
- only one
- only two
- only three

Concept

Made by: Waqar Siddhu

Question No : 6 of 52 Marks: 1 (Budgeted Time 1 Min)

Even-palindrome is a _____ language

Answer (Please select your correct option)

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- non-regular
- regular
- regular but infinite
- regular but finite

Lec:25,page:71

Made by: Waqar Siddhu

Question No : 7 of 52 Marks: 1 (Budgeted Time 1 Min)

If L1 and L2 are two regular languages, then _____ is/are also regular.

Answer (Please select your correct option)

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- Both L1 ∪ L2 , L1 ∩ L2 lec 24,page 65 L1+L2= L1UL2 and on page 67
- L1' ∩ L2
- L1 ∪ L2
- L1 ∩ L2

Made by: Waqar Siddhu

Question No : 8 of 52 Marks: 1 (Budgeted Time 1 Min)

If the FA has N states, then test the words of length less than N. If any one word is accepted by this FA, then it will accept ____ word.

Answer (Please select your correct option)

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- All Lec:30,page:85 concept
- No
- One
- Two

Made by: Waqar Siddhu

Question No : 9 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following pairs of regular expressions are equivalent?

Answer (Please select your correct option)

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- 1(001)* and (10)*10
- x(xx)* and (x)*x concept
- x^+ and x^*
- x^+ and x^*x^+

Made by: Waqar Siddhu

Question No : 10 of 52 Marks: 1 (Budgeted Time 1 Min)

An alphabet of Σ is valid if

Answer (Please select your correct option)

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- ○ No letter of Σ appears in middle of any other letter
- ○ No letter of Σ appears at end of any other letter
- ● No letter of Σ appears at start of any other letter

Lec:1,page:4

- ○ No letter of Σ appears at end or middle of any other letter

Made by: Waqar Siddhu

Question No : 11 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following pairs of regular expressions are not equivalent?

Answer (Please select your correct option)

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- ○ (ab)* and a*b*
- ○ x(xx)* and (xx)*x
- ● x^+ and x^*x^+

Concept

- ○ x(x)* and (x)*x

Made by: Waqar Siddhu

Question No : 12 of 52 Marks: 1 (Budgeted Time 1 Min)

There exist _____ derivations of a certain word in ambiguous grammar.

Answer (Please select your correct option)

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- ● more than one
- ○ Exactly one
- ○ exactly two
- ○ no

Made by: Waqar Siddhu

Question No : 13 of 52 Marks: 1 (Budgeted Time 1 Min)

Regular languages are closed under _____

Answer (Please select your correct option)

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- ● Union and Concatenation
- ○ Union and Division
- ○ Union and Exponent
- ○ Union and Factorial

Concept

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Question No : 14 of 52 Marks: 1 (Budgeted Time 1 Min)

Which one of the following statements is INCORRECT

Answer (Please select your correct option)

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- ● A language consisting of all strings over ∑=(a,b) having equal number of a's and b's is a regular language
- ○ Every language that can be expressed by FA can also be expressed by RE
- ○ None of these
- ○ (a+b)*aa(a+b)* generates Regular language

Lec:26,page:73

Made by: Waqar Siddhu

Question No : 15 of 52 Marks: 1 (Budgeted Time 1 Min)

TM is more powerful than FSM because

Answer (Please select your correct option)

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- ○ The tape movement is confined to one direction
- ○ It has no finite state control
- ○ It has the capability to remember arbitrary long sequences of input symbols
- ● The tape movement is confined to more than one directions

Made by: Waqar Siddhu

Question No : 16 of 52 Marks: 1 (Budgeted Time 1 Min)

CYK algorithm was invented by John Cocke and later was published by ____

Answer (Please select your correct option)

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- Talik Khon and Daniel H
- Tandao Kasami and Daniel H
- Tandao Dagger and Daniel H
- Talik Khon

lec:44,page:135

Made by: Waqar Siddhu

Question No : 17 of 52 Marks: 1 (Budgeted Time 1 Min)

The process of finding the derivation of word generated by particular grammar is called_____

Answer (Please select your correct option)

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- Processing
- Parsing
- Programming
- Planing

Lec:44,page:136

Made by: Waqar Siddhu

Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

There are different parsing techniques, including _____

Answer (Please select your correct option)

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- Top down parsing
- Bottom up parsing
- Parsing technique for particular grammar of arithmetic expression
- All of these

Lec:44,page:136

Made by: Waqar Siddhu

Question No : 19 of 52 Marks: 1 (Budgeted Time 1 Min)

Σ=(a,Aa,Abb) , then string aAaAbbAa has length equal to_____.

Answer (Please select your correct option)

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- Three
- Four
- One
- Two

concept

Made by: Waqar Siddhu

Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

The language generated by CFG is called _____.

Answer (Please select your correct option)

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- Infinite
- Context Free Language (CFL)
- Regular
- Finite

Lec:31,page:88

Made by: Waqar Siddhu

Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

In _____ there must be transition for all letters.

Answer (Please select your correct option)

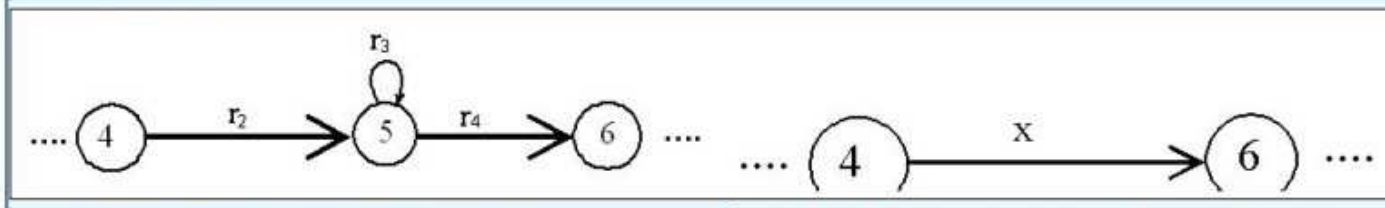
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- NFA
- FA
- TG
- GTG

Lec:4,page:11

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Question No : 22 of 52 Marks: 1 (Budgeted Time 1 Min)



Answer (Please select your correct option)

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- r1* r2 r3*
- (r1r2r3)*
- r1*r2*r3*
- None of the given incomplete Q..RE of ist is r2r3*r4

Made by: Waqar Siddhu

Question No : 23 of 52 Marks: 1 (Budgeted Time 1 Min)

A regular language _____ .

Answer (Please select your correct option)

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- Must be finite
- Must be infinite
- Can be finite or infinite concept
- Complex Language

Made by: Waqar Siddhu

Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

If a language can be expressed by a regular expression, then its complement is _____ .

Answer (Please select your correct option)

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- Regular Lec:24,page:66
- Non Regular
- Finite
- Infinite

Made by: Waqar Siddhu

Question No : 25 of 52 Marks: 1 (Budgeted Time 1 Min)

For every Context Free Grammar (CFG), we can make the corresponding _____.

Answer (Please select your correct option)

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- FA
- TG
- PDA

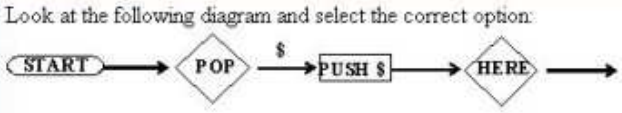
Lec:39,page:111

- Regular Grammar

Made by: Waqar Siddhu

Question No : 26 of 52 Marks: 1 (Budgeted Time 1 Min)

Look at the following diagram and select the correct option:



Answer (Please select your correct option)

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- The PDA must not begin with the sequence
- The PDA may begin with the sequence
- The PDA must begin with the sequence

Lec:40,page:119

- The PDA may not begin with the sequence

Made by: Waqar Siddhu

Question No : 27 of 52 Marks: 1 (Budgeted Time 1 Min)

Consider the following Context Free Grammar (CFG) and select the correct choice
S→AB
A →BSB | a | b
B→CC
C→SS | b | bb

Answer (Please select your correct option)

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- abb is a word in the corresponding CFL.

concept

- abb is not the word of corresponding CFL.
- Any word can be accepted from the corresponding CFL.
- b is a word in the corresponding CFL.

Made by: Waqar Siddhu

Question No : 28 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following statement(s) is/are true or false?

1) The turing machine is similar to a finite automaton but with an unlimited and unrestricted memory.
2) A Turing machine is a much more accurate model of a general purpose computer.

Answer (Please select your correct option)

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- (•) Statement 1 is true — Mentioned in lec 45
- ( ) Statement 2 is true — Both are true..check link in moazz file
- ( ) Both statements (1 & 2) are false
- ( ) Statement 2 is false

Made by: Waqar Siddhu

Question No : 29 of 52 Marks: 1 (Budgeted Time 1 Min)

Language of strings having even number of a's is considered to be:

Answer (Please select your correct option)

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- (•) Regular — Lec:3,page:9
- ( ) Irrelgular
- ( ) Irregular and infinite
- ( ) Irregular and finite

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Question No : 30 of 52 Marks: 1 (Budgeted Time 1 Min)

If we have two transiton graphs then their union will be expressed by:

Answer (Please select your correct option)

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- (•) taking a common start state and joining them by two null transitions — Lec:24,page:65
- ( ) just connecting both start states by null transitions
- ( ) connecting final state of first TG to the initial state of second TG
- ( ) connecting the final state of first TG to the final state of second TG

Made by: Waqar Siddhu

Question No : 31 of 52 Marks: 1 (Budgeted Time 1 Min)

A language of strings of prime length is considered to be:

Answer (Please select your correct option)

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- regular but infinite
- regular
- irregular Lec:27,page:75
- regular but finite

Made by: Waqar Siddhu

Question No : 32 of 52 Marks: 1 (Budgeted Time 1 Min)

_____ and _____ are removed in order to make a CFG in Chomsky Normal Form (CNF).

Answer (Please select your correct option)

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- Null, nullable productions
- Nullable, unit productions
- Null, unit productions Lec:36,page:102
- Strings of length 0, null

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Question No : 33 of 52 Marks: 1 (Budgeted Time 1 Min)

The languages are supposed to be _____ in case of testing pumping lemma.

Answer (Please select your correct option)

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- context free
- irregular
- both regular and irregular
- regular mentioned in video lec: 27 :P can check exm

Made by: Waqar Siddhu

Question No : 34 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following refers to an effectively solvable problem having answer in "Yes" or "No"?

Answer (Please select your correct option)

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- ● Decision procedure

Lec:29,page:80

- ○ Logical problem
- ○ Analytical problem
- ○ Binary procedure

Made by: Waqar Siddhu

Question No : 35 of 52 Marks: 1 (Budgeted Time 1 Min)

Consider the following Context Free Grammar (CFG):

S ---> SX|XY
X ---> ZZ
Y ---> 0
Z ---> 1

Answer (Please select your correct option)

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- ● 110
- ○ 11011
- ○ 1101111
- ○ 11101

Made by: Waqar Siddhu

Question No : 36 of 52 Marks: 1 (Budgeted Time 1 Min)

Consider the following Context Free Grammar (CFG):

S → SX|XY
X → ZZ
Y → 1
Z → 0

Answer (Please select your correct option)

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- ● 001
- ○ 00010
- ○ 0010000
- ○ 00100

Made by: Waqar Siddhu

Question No : 37 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is a regular Context Free Grammar:

Answer (Please select your correct option)

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- ● S→abS | baS | ^
- ○ S→aSb | baS |^
- ○ S→abS | bSa | ^
- ○ S→aSb | Sa | ^

Made by: Waqar Siddhu

Question No : 38 of 52 Marks: 1 (Budgeted Time 1 Min)

Concatenation of r1 and r2 will be shown as:

Answer (Please select your correct option)

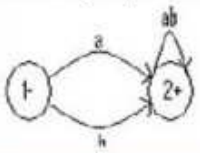
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- ○ r1+r2
- ○ r2+r1
- ○ r2r1
- ● r1r2 Lec:13,page:35

Made by: Waqar Siddhu

Question No : 19 of 26 Marks: 1 (Budgeted Time 1 Min)

How many steps are required to convert the given transition graph into a regular expression?



Answer (Please select your correct option)

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- ○ 1 step
- ● 2 steps firstly a & b combine to form a+b and in 2nd step loop of ab combine =(a+b)ab*
- ○ 3 steps
- ○ 4 steps

Made by: Waqar Siddhu

Question No : 39 of 52 Marks: 1 (Budgeted Time 1 Min)

A nonterminal is said to be self-embedded, if in a given derivation of a word, it ever occurs as a tree _____ of itself.

Answer (Please select your correct option)

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- ascendant
- descendant
- top
- bottom

Lec:43,page:127

Made by: Waqar Siddhu

Question No : 40 of 52 Marks: 1 (Budgeted Time 1 Min)

_____ is always Deterministic.

Answer (Please select your correct option)

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- Finite Automaton
- Transition Graph
- Generalize Transition Graph
- Non-deterministic finite automaton

lec:10,page:25

Made by: Waqar Siddhu

Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

Write alternative form of this production.
$S \rightarrow aS, S \rightarrow bS, S \rightarrow \Lambda$

Answer (Please click here to Add Answer)

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S ⟶ aS|bS|Λ

e:90

Made by: Waqar Siddhu

Question No : 42 of 52 Marks: 2 (Budgeted Time 4 Min)

What are the uses of Push Down Automata (PDA) in computing?

Answer (Please click here to Add Answer)

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PUSHDOWN STACK is a place where the input letters can be placed until these letters are referred again.
Itcan store as many letters as one can in a long column

Made by: Waqar Siddhu

Question No : 43 of 52 Marks: 2 (Budgeted Time 4 Min)

What is meant by Row language?

Answer (Please click here to Add Answer)

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Lec:41,page:123

Made by: Waqar Siddhu

Question No : 44 of 52 Marks: 2 (Budgeted Time 4 Min)

Write down regular expression for a language L of strings defined over $\Sigma = (a, b)$ that neither ends in "a" nor with "b".

Answer (Please click here to Add Answer)

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RE=^ (null)

Made by: Waqar Siddhu

Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

What are the steps of Recursive Definition of Languages?

Answer (Please click here to Add Answer)

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Lec:3,page:9

Made by: Waqar Siddhu

Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

What does the following arbitrary summary table show?

FROM Where	TO Where	READ What	POP What	PUSH What	ROW Number
$READ_0$	$READ_1$	b	b	abb	11

Answer (Please click here to Add Answer)

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Lec:42,page:124

Made by: Waqar Siddhu

Question No : 47 of 52 Marks: 3 (Budgeted Time 6 Min)

Write down Context Free Grammar (CFG) for the language of strings expressed by the following regular expression:

$(a + b)^*(a + b)^*$

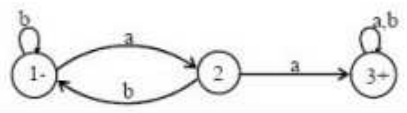
Answer (Please click here to Add Answer)

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Question No : 48 of 52 Marks: 3 (Budgeted Time 6 Min)

Let us consider the following Finite Automaton (FA):



Answer (Please click here to Add Answer)

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Lec:6,page:16

Made by: Waqar Siddhu

Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

Write the CFG for EVEN-EVEN language?

Answer (Please click here to Add Answer)

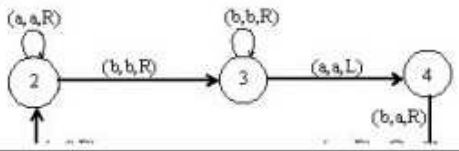
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Lec:32,page:90

Made by: Waqar Siddhu

Question No : 50 of 52 Marks: 5 (Budgeted Time 10 Min)

Consider the following TM



Answer (Please click here to Add Answer)

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page 142

Made by: Waqar Siddhu

Question No : 51 of 52 Marks: 5 (Budgeted Time 10 Min)

Using suitable examples, differentiate between Kleene star closure and plus operation?

Answer (Please click here to Add Answer)

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Lec:2,page:7

Made by: Waqar Siddhu

Question No : 52 of 52 Marks: 5 (Budgeted Time 10 Min)

a. Construct Context Free Grammar (CFG) for the language defined by the regular expression:

(ab + ba)*(a + b)*

b. Remove null productions from the CFG constructed in part i.

Answer (Please click here to Add Answer)

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Theory of Automata(Cs402)
Final term Papers
Solved final term by Hina

Question: 1
The production of the form nonterminal →Λ is said to be null production.
- Ø ► True Page 104
- Ø ► False

Question: 2
Consider the following GFC :
s→aa|bB,a→aa|B,B→aS|Λ
here s→aaand A→B are null productions ,while B→Λ is null able production
- Ø ►false page 105
- Ø ►True

Question: 3
Which statement is true?
- Ø The tape of turing machine is infinite.
- Ø The tape of turing machine is finite.
- Ø The tape of turing machine is infinite when the language is regular
- Ø The tape of turing machine is finite when the language is nonregular.

Question: 4
The production of the form nonterminal →one nonterminal is called the :
- Ø Null production
- Ø Null able production
- Ø Unit production page105
- Ø None of the given

Question: 5
For a given input, it provides the compliment of Boolean AND output.
- Ø NAND box (NOT AND)
- Ø DELAY box
- Ø OR box
- Ø AND box

Question: 6
(Marks: 1) - Please choose one
Choose the correct statement.

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Ø A Mealy machine generates no language as such Ø A Moore machine generates no language as such Ø **A Mealy machine has no terminal state**

Ø All of these

Question: 7

Let Q and R be expressed by ab*a and (ba)* respectively i.e Q={aa ,aba ,abba ……}and R={Λ,ba baba,bababa…….}..aba is the only word in Q which can make a word in R, because the words in R don,t contain the

Ø **Single letter**

Ø **Double letter** **page 84**

Ø String

Ø Null string

Question: 8

It delays the transmission of signal along the wire by one step (clock pulse).

Ø OR box

Ø **DELAY box**

Ø NAND box (NOT AND)

Ø AND box

Question: 9

To describe the complement of a language, it is very important to describe the ---------- of that language over which the language is defined.

Ø String

Ø Regular Expression

Ø **Alphabet**

Ø Word

Question:10

Let L be a language defined over an alphabet ∑,then the language of strings , defined over ∑, not belonging to L denoted by L^C or L. is called :

Ø Non regular language of L

Ø **Complement of the language L**

Ø Non of the given

Ø All of above

Question:11

For the given input, it provides the Boolean OR output

Ø DELAY box

Ø AND box

Ø NAND box (NOT AND)

Ø **OR box**

Question: 12

For the given input, AND box provides the Boolean AND output.

Ø **True**

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- Ø False

Question: 13
The current in the wire is indicated by 1 and 0 indicates the absence of the current.
- Ø **True**
- Ø False

Question: 14
If L is a regular language ,then according to Kleene,theorem , there exists an :
- Ø TG
- Ø GTG
- Ø **FA page77**
- Ø Non of the given

Question: 15
Any language that can not be expressed by a RE is said to be regular language.
- Ø True
- Ø **False**

Question: 16
and r2 = (a + b) then the language (aa + bb)(a + b) will be generated by

- Ø **(r1 + r2) Ø**
- (r2)(r1) Ø
- (r1)*
- Ø (r1)(r2)

Question: 17
If L1 and L2 are regular languages is/are also regular language(s).

- Ø L1 + L2
- Ø L1L2
- Ø L1*
- Ø **All of above Page 70**

Question: 18
Let L be a language defined over an alphabet Σ, then the language of strings, defined over Σ, not belonging to L, is called Complement of the language L, denoted by Lc or L'.
True
False

NO19: For a certain language L, the complement of L^C is the given language L *i.e.* $(L^C)^C = L$
- Ø **True Page 71**
- Ø **False**

Question: 20
If L is a regular language then, L^C is also a _____language.

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- Ø Regular
- Ø Non-regular
- Ø Regular but finite
- Ø None of the given

Question: 21

If an effectively solvable problem has answered in yes or no, then this solution is called ---------

- Ø Decision problem
- Ø Decision method
- Ø Decision procedure
- Ø Decision making

Question: 22

There is an approach in defining the quotient of regular languages ie the language Q is said to be quotient of two regular languages P and R, denoted by Q=R/P if :

- Ø PQ=R Ø

R=PQ Ø

QR=P

- Ø Non of above

Question: 23

consider a language L defined over an alphabet ∑ if two strings x and y defined over ∑ are run over an FA accepting the language L,then x and y are said to belong to the same ……. if they end in the same …….

- Ø Class ,state
- Ø Final ,infinite
- Ø Regular ,nonregular
- Ø All of Above

Question:24

If L is a regular language then, --------- is also a regular language.

- Ø Lm Ø

Ls Ø Lx

- Ø Lc

Question:25

Converting each of the final states of F to non-final states and old non-final states of F to final states, FA thus obtained will reject every string belonging to L and will accept every string, defined over Σ, not belonging to L. is called

- Ø Complement of L
- Ø Finite Automata of L Ø

Transition Graph of L Ø

Regular expression of L

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Question: 26

If L1 and L2 are two regular languages, then L1 U L2 is not a regular.

- Ø **True**
- Ø **False**

Question:27

L= language of words containing even number of a's. Regular Expression is

- Ø **(a+b)*aa(a+b)***
- Ø **(b+ab*a)***
- Ø **a+bb*aab*a**
- Ø **(a+b)*ab(a+b)***

Question:28

The regular expression defining the language L1 U L2 can be obtained, converting and reducing the previous ---------- into a ---------- as after eliminating states.

- Ø **GTG, TG** Ø
- **FA, GTG** Ø **FA, TG**
- Ø **TG, RE**

Question: 29

The language that can be expressed by any regular expression is called a Non regular language.

True

False Page 76

Question:30

Please choose one

Choose the incorrect statement:

- Ø **(a+b)*aa(a+b)* generates Regular language.**
- Ø **A language consisting of all strings over ∑={a,b} having equal number of a's and b's is a regular language**
- Ø **Every language that can be expressed by FA can also be expressed by RE**
- Ø **None of these**

Question: 31

The languages ---------- are the examples of non regular languages

- Ø **EVEN-EVEN and PRIME**
- Ø **PALINDROME and PRIME. Page 76**
- Ø **PALINDROME and EVEN-EVEN**
- Ø **FACTORIAL and SQURE**

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Question: 32

De-Morgan's law for sets is expressed by,

- Ø $(L_1^c \cap L_2^c)^c = L_1^c \cap L_2^c$
- Ø $(L_1^c \cap L_2^c)^c = L_1 \cap L_2$
- Ø $(L_1^c \cap L_2^c)^c = L_1 \cup L_2$
- Ø $(L_1^c \cap L_2^c)^c = L_1^c \cap L_2^c$

Question: 33

Let L be any infinite regular language, defined over an alphabet Σ then there exist three strings x, y and z belonging to Σ* such that all the strings of the form $xy^n z$ for n=1,2,3, … are the words in L. called Complement of L

- Ø **Pumping Lemma** **Page 77**
- Ø **Kleene's theorem**
- Ø **None of the given**
- Ø **1,2 both**

Question: 34

Languages are proved to be regular or non regular using pumping lemma.

- Ø **True**
- Ø **False**

Question:35

---------- is obviously infinite language.

- Ø **EQUAL-EQUAL**
- Ø **EVEN-EVEN**
- Ø **PALINDROME** **Page 80**
- Ø **FACTORIAL**

Question: 36

If L1 and L2 are expressed by regular expressions r1 and r2, respectively then the language expressed by r1 + r2 will be _____

- Ø **Ir-regular**
- Ø **Can't be decided**
- Ø **Regular language** **Page 77**
- Ø **Another Language which is not listed here**

Question: 37

Let L be an infinite language accepted by a language accepted by a finite automaton with N states , then for all words W in L that have length more than N there are strings x,y and Z (y being non null string)and length (x)+length (y)≤Ns.t.W=xyz and all strings of the form $xy^n z$ are in L for n=1,2,3…….

- Ø **True** **Page 80**

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Ø False

Question: 38

If, two strings x and y, defined over Σ, are run over an FA accepting the language L, then x and y are said to belong to the same class if they end in the same state, no matter that state is final or not.

Ø True

Ø False

Question: 39

Myhill Nerode theorem is consisting of the followings.

Ø L partitions Σ* into distinct classes.

Ø If L is regular then, L generates finite number of classes. Ø If L generates finite number of classes then L is regular. Ø All of above

Page 80

Question:40

The language Q is said to be quotient of two regular languages P and R, denoted by--- if PQ=R.

Ø R=Q/P

Ø Q=R/P Page 83

Ø Q=P/R

Ø P=R/Q

Question:41

If two languages R and Q are given, then the prefixes of Q in R denoted by Pref(Q in R).

Ø True Page 83

Ø False

Question:42

Let Q = {aa, abaaabb, bbaaaaa, bbbbbbbbbb} and R = {b, bbbb, bbbaaa, bbbaaaaa} Pref (Q in R) is equal to

Ø {b,bbba,bbbaaa}, Page 83

Ø {b,bba,bbaaa} Ø

{ab,bba,bbbaa} Ø

{b,bba,bbba}

Question: 43

If R is regular language and Q is any language (regular/ non regular), then Pref (Q in R) is ---------.

Ø Non-regular

Ø Equal

Ø Regular Page 84

Ø Infinite

Question: 44

The regular expression thus obtained if contains at least one word then the language is not empty

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otherwise the language is :

- Ø **Regular**
- Ø **No regular**
- Ø **Empty** **Page 85**
- Ø **Non of the above**

Question: 45
"CFG" stands for _____:

- Ø **Context Free Graph**
- Ø **Context Free Grammar** **Page 92**
- Ø **Context Finite Graph**
- Ø **Context Finite Grammar**

Question:46
The langue generated by CFG is called Context free language (CFL)

- Ø **False**
- Ø **True**

Question:47

_____states are called the halt states.

- Ø **ACCEPT and REJECT**
- Ø **ACCEPT and READ**
- Ø **ACCEPT AND START**
- Ø **ACCEPT AND WRITE**

Question:48
If a regular expression contains * then it may define an infinite language ,with exception Λ*as Λ*= Λ e.g.

- Ø **(Λ+a Λ*)(Λ*+Λ) * defines finite language. While (Λ+a Λ*)*(Λ*+Λ) *defines an finite language.**
- Ø **True**
- Ø **False** **Page 90**

Question:49
The part of an FA, where the input string is placed before it is run, is called

- Ø State
- Ø Transition
- Ø **Input Tape** **Page 110**
- Ø Output Tape

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Question: 50

TM is more powerful than FSM because

- Ø The tape movement is confined to one direction
- Ø **It has no finite state control**
- Ø It has the capability to remember arbitrary long sequences of input symbols
- Ø None of these

NO51: In new format of an FA This state is like dead-end non final state:

- Ø ACCEPT
- Ø **REJECT Page 110**
- Ø STATR
- Ø READ

NO52: For language L defined over {a, b}, then L partitions {a, b}* into …… classes

- Ø Infinite
- Ø Finite
- Ø **Distinct**
- Ø Non-distinct

NO53 := language of words containing even number of a's. Regular Expression is

- Ø (a+b)*aa(a+b)*
- Ø **(b+ab*a)***
- Ø a+bb*aab*a
- Ø (a+b)*ab(a+b)*

No 54:All NonNull words of the CFL can be generated by the corresponding CFG which is in CNF i.e the grammar in CNF will generate the same language except the :

- Ø string
- Ø regular language
- Ø **null string .**
- Ø non of above

NO55: The ……. is said to be ambiguous if there exist at least one word of its language that can be generated by the different production tree .

- Ø CFL
- Ø CFG Page 98
- Ø GTG
- Ø None of the given

NO56 : Between the two consecutive joints on a path

- Ø One character can be pushed and one character can be popped
- Ø **Any no. of characters can be pushed and one character can be popped**
- Ø One character can be pushed and any no. of characters can be popped
- Ø Any no. of characters can be pushed and any no. of characters can be popped

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NO57: In pumping lemma theorem (x y^n z) the range of n is:

- Ø **n=1, 2, 3, 4…….** **Page 77**
- Ø n=0, 1, 2, 3, 4…….
- Ø n=…….-3,-2,-1, 0, 1, 2, 3, 4……
- Ø n=…….-3,-2,-1, 1, 2, 3, 4……

NO58: TM is more powerful than FSM because Ø

The tape movement is confined to one direction Ø **It has no finite state control**

- Ø It has the capability to remember arbitrary long sequences of input symbols
- Ø None of these

NO59 : If every production in CFG is one of the following forms
Conterminal → semi word
Nonterminal→word
Then the language generated by that GFC is :

- Ø **Regular**
- Ø Nonregular
- Ø Finite
- Ø Infinite

NO 60: Then the language generated by that CFG is:

- Ø Non regular
- Ø Infinite
- Ø **Regular** **Page 102**
- Ø Finite

NO 61:The PDA is called non-deterministic PDA when there are more than one out going edges from…… state :

- Ø START or READ
- Ø POP or REJECT
- Ø **READ or POP** **Page 116**
- Ø PUSH or POP

NO:62 Identify the TRUE statement:

- Ø A PDA is non-deterministic, if there are more than one READ states in PDA
- Ø A PDA is never non-deterministic
- Ø **Like TG, A PDA can also be non-deterministic** **Page 116**
- Ø A PDA is non-deterministic, if there are more than one REJECT states in PDA

NO:63 the language Q is infinite.

- Ø **True** **Page 134**
- Ø **False**

Left hand side of a production in CFG consists of:

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- Ø One terminal
- Ø More than one terminal
- Ø One non-terminal
- Ø Terminals and non-terminals Page 64:

NO65: it is very important to determine which sequences of rows do correspond to possible paths through the:

- Ø CFG
- Ø CFL
- Ø **PDA Page 128**
- Ø TAPE

NO: 66 : The following problem(s) ---------- is/are called decidable problem(s).

- Ø The two regular expressions define the same language
- Ø The two FAs are equivalent
- Ø **Both a and b**
- Ø None of given

NO67: Before the CFG corresponding to the given PDA is determined,the PDA is converted in to the standard form which is called the:

- Ø Standard form
- Ø **Conversion form Page 108**
- Ø Left most derivation
- Ø None of them

NO: 68 The deviation of the word W generated by a CFG, such that at each step ,a production is applied to the left most nonterminal in the working string is said to be

- Ø Left most terminal
- Ø **Left most deviation : Page 108**
- Ø None of these
- Ø A ,B both

NO 69: To examine whether a certain FA accepts any words, it is required to seek the paths from ----- -- state.

- Ø Final to initial
- Ø Final to final
- Ø **Initial to final**
- Ø Initial to initial

NO 70: The high level language is converted into assembly language codes by a program called compiler.

- Ø **TRUE**
- Ø FALSE

NO 71: Grammatical rules which involve the meaning of words are called ----------

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- Ø **Semantics** **Page 92**

- Ø Syntactic Both
- Ø a and b None
- Ø of given

NO72: Choose the correct statement.

A Mealy machine generates no language as such
A Moore machine generates no language as such
A Mealy machine has no terminal state
All of these

NO: 73 Grammatical rules which do not involve the meaning of words are called ----------

- Ø Semantics
- Ø **Syntactic** **Page 92**

- Ø **Both a and b**
- Ø **None of given**

NO74: - Please choose one
The word 'formal' in formal languages means

- Ø They are unnecessary, in reality
- Ø Only the form of the string of symbols is significant
- Ø **The symbols used have well defined meaning**
- Ø None of these

NO: 75 ……. is a place where the input letters can be placed until these letters are referred again. It can store as many letters as one can in a long column .

- Ø STACK
- Ø POP AND STACK
- Ø **PUSHDOWN STACK** **Page 112**
- Ø **None of above**

NO 76: Consider the language L of strings, defined over Σ = {a,b}, ending in a

► There are finite many classes generated by L, so L is regular
► **There are infinite many classes generated by L, so L is regular**
► There are finite many classes generated by L, so L is non-regular
► There are infinite many classes generated by L, so L is non-regular

NO77: The symbols that can't be replaced by anything are called ----------

- Ø Productions
- Ø **Terminals Page 92**
- Ø Non-terminals
- Ø All of above

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NO78: "One language can be expressed by more than one FA". This statement is _____

- Ø **True**
- Ø False
- Ø Some times true & sometimes false
- Ø None of these

NO: 79 The symbols that must be replaced by other things are called _____

- Ø Productions
- Ø Terminals
- Ø **Non-terminals** **Page 92**
- Ø None of given

NO 80: Which of the following statement is NOT true:

- Ø FA can be considered to be an NFA
- Ø FA can be considered to be an NFA with null string
- Ø NFA can be considered to be an TG
- Ø **TG can be considered to be an NFA**

NO 81: Let FA 3 be an FA accepting L1∩L2 then the initial stat of FA3 must be correspond to the initial state of……. and initial state of …… p 74

- Ø FA3, FA2
- Ø **FA1 , FA2**
- Ø FA1,FA3
- Ø None of the given

NO82: If r1 = (aa + bb) and r2 = (a + b) then the language (aa + bb)(a + b) will be generated by

- Ø (r1)(r2)
- Ø **(r1 + r2)**
- Ø (r2)(r1) Ø
  (r1)*

NO83: The grammatical rules are often called_____

- Ø **Productions** **Page 92**
- Ø Terminals
- Ø Non-terminals
- Ø None of given

NO 84: Does the empty string match the regular expression |y+a|?

- Ø Yes
- Ø **No**

NO 85
The terminals are designated by _____letters, while the non-terminals are designated by _____letters.

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- Ø Capital, bold
- Ø **Small, capital** **Page 92**
- Ø Capital, small
- Ø Small, bold

NO86: FA corresponding to an NFA can be built by introducing a state corresponding to the combination of states, for a letter having
Choices:

- Ø **no transition at certain state**
- Ø **one transition at certain state**
- Ø **more than one transitions at certain state**
- Ø **none of the given options**

NO:87 Σ = {a,b} Productions S→XaaX X→aXX→bX X→Λ This grammar defines the language expressed by_____

- Ø **(a+b)*aa(a+b)*** **Page 94**
- Ø (a+b)*a(a+b)*a
- Ø (a+b)*aa(a+b)*aa
- Ø (a+b)*aba+b)*

NO 88: Which statement is true:

- Ø ► **The tape of turing machine is infinite.**
- Ø ► The tape of turing machine is finite.
- Ø ► The tape of turing machine is infinite when the language is regular
- Ø ► The tape of turing machine is finite when the language is nonregular.

NO89 : The language generated by _____ is called Context Free Language (CFL).

- Ø FA
- Ø TG
- Ø **CFG** **Page 93**
- Ø TGT

NO 90: Let A = {0, 1}. The number of possible strings of length 'n' that can be formed by the elements of the set A is:

- Ø n!
- Ø n^2
- Ø n^m
- Ø 2^n

NO : 91 S → aXb|bXa X → aX|bX|Λ The given CFG generates the language of strings in English _____

- Ø **Beginning and ending in different letters** **Page 96**
- Ø Beginning and ending in same letter

- Ø Having even-even language
- Ø None of given

NO92: Every regular expression can be expressed as CFG but every CFG cannot be expressed as a regular expression. This statement is:

- Ø Depends on the language
- Ø None of the given options
- Ø **True**
- Ø False

NO 93: The CFG is said to be ambiguous if there exists atleast one word of its language that can be generated by the different production trees,

- Ø **TRUE Page 100**
- Ø FALSE

NO 94: The language generated by that CFG is regular if _____

- Ø No terminal → semi word
- Ø No terminal → word
- Ø **Both a and b Page 102**
- Ø None of given

NO95: A regular language:

- Ø **Must be finite**
- Ø Must be infinite
- Ø Can be finite or infinite
- Ø Must be finite and cannot be infinite

NO96: The production of the form non terminal → Λ is said to be null production .

- Ø **TRUE Page 104**
- Ø **FALSE**

NO 97: Who did not invent the Turing machine?

- Ø Alan Turing
- Ø **A. M. Turing**
- Ø Turing
- Ø None of these

NO: 98 A production is called null able production if it is of the form N → Λ

- Ø **TRUE Page 105**
- Ø FALSE

NO99: A DFA with n states must accept at least one string of length greater than n.

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Choices:

- Ø **True**
- Ø False

NO100: For every three regular expressions R, S, and T, the languages denoted by R(S U T) and (RS) U (RT) are the same.

Choices:

- Ø **True**
- Ø False

NO101: Choose the right option: (2)

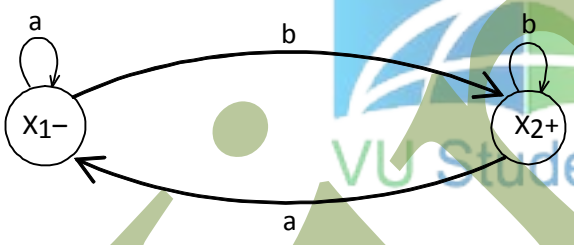
In a Mealy machine, the set of letters and the set of output characters must be same

In a Mealy machine, the set of letters and the set of output characters may not be same

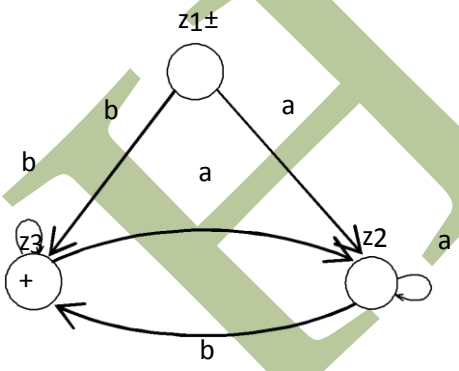
B only

A only

No102 : - Please choose one



Above given FA corresponds RE r. then FA corresponding to r* will be



This statement is

True **Page 42**

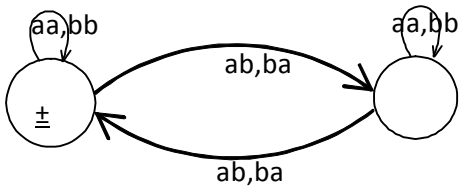
False

Depends on language

None of these

NO103 : - Please choose one

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Above given TG has _____RE.

(aa+aa+(ab+ab)(aa+ab)*(ab+ba))*

(aa+bb+(ab+ba)(aa+bb)*(ab+ba))* Page 26

(aa+bb+(ab+ba)(aa+bb)(ab+ba))*

None of these

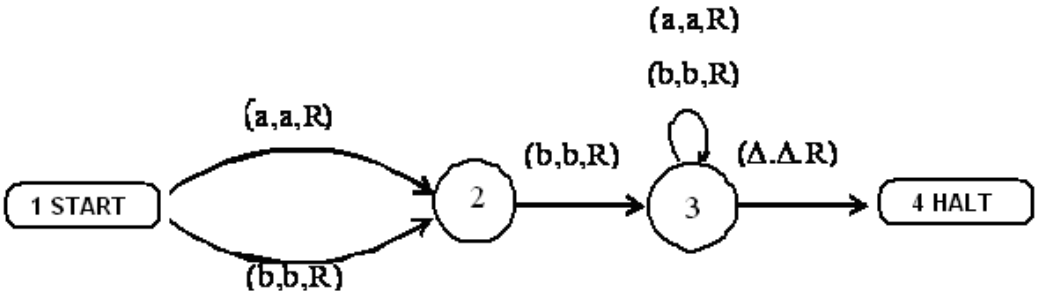
NO104: - Please choose one

Like TG, a PDA can also be non-deterministic

True

False

NO105 :- Please choose one :



The above machine is a/anTG _____http://vustudents.ning.com

Finite Automata

Turing machine Page 148

FA

TG

NO106: In FA, if one enters in a specific state but there is no way to leave it, then that specific state is called :

Dead State Waste

Basket Davey John

Locker All of these

NO107: - Please choose one

In CFG, the symbols that can't be replaced by anything are called_____

Ø Terminal

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- Ø Non-Terminal
- Ø **Production**
- Ø All of given

NO108: - Please choose one
Which of the following is NOT a regular language?

- Ø String of 0's whose length is a perfect squere
- Ø **Set of all palindromes made up of 0's and 1's**
- Ø String of 0's whose length is a prime number
- Ø All of the given options

NO109: - Please choose one
Choose the incorrect (FALSE) statement

- Ø A Mealy machine generates no language as such
- Ø A Mealy machine has no terminal state
- Ø **For a given input string, length of the output string generated by a Moore machine is not more than the length of the output string generated by that of a Mealy machine . Page 62**
- Ø All of these

NO110: - Please choose one
Pumping lemma is generally used to prove that

- ► A given language is infinite
- ► **A given language is not regular : Page 77**
- ► Whether two given regular expressions of a regular language are equivalent or not
- ► None of these

NO111:- Please choose one
Which of the following is a regular language?

- Ø String of odd number of zeroes
- Ø Set of all palindromes made up of 0's and 1's
- Ø **String of 0's whose length is a prime number**
- Ø All of these

NO112 : lemma is generally used to prove that

- Ø A given language is infinite
- Ø **A given language is not regular Page 78**
- Ø Whether two given regular expressions of a regular language are equivalent or not
- Ø None of these

NO113: language can be expressed by more than one FA". This statement is _____

- Ø **True**
- Ø False
- Ø Some times true & sometimes false
- Ø None of these

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NO114: language:

- Must be finite
- **Must be infinite**
- Can be finite or infinite
- Must be finite and cannot be infinite

NO115 : enters in a specific state but there is no way to leave it, then that specific state is called

- Dead State
- Waste Basket
- Davey John Locker
- **All of these**

NO116: symbols that can't be replaced by anything are called_____

- **Terminal** **Page 92**
- Non-Terminal
- Production
- All of given

NO117: following is NOT a regular language?

- String of 0's whose length is a perfect squere
- Set of all palindromes made up of 0's and 1's Ø
  String of 0's whose length is a prime number Ø
  All of the given options

NO118: Left hand side of a production in CFG consists of

- One terminal
- More than one terminal
- One non-terminal
- **Terminals and non-terminals** **Page 92**

NO119: One language can be expressed by more than one FA". This statement is _____

- **True**
- False
- Some times true & sometimes false
- None of these
-

NO120: invent the Turing machine?

- Alan Turing
- **A. M. Turing**
- Turing
- None of these

'NO121: formal' in formal languages means

- **The symbols used have well defined meaning**
- They are unnecessary, in reality
- Only the form of the string of symbols is significant

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- Ø None of these

NO122: TM is more powerful than FSM because

- Ø The tape movement is confined to one direction
- Ø **It has no finite state control**
- Ø It has the capability to remember arbitrary long sequences of input symbols
- Ø None of these

NO123: A program which is the set of rules which show that which state is to be entered when a letter is read form the :

- Ø **TAPE** **Page 147**
- Ø HALT
- Ø TM
- Ø None of above

NO124: The process of finding the derivation of word generated by particular grammar is called :

- Ø PLUS TIMING
- Ø **Parsing** **Page 142**
- Ø HALT
- Ø All of above

NO125: For a non regular language there exist …… FA:

- Ø **NO**
- Ø Yes

NO126: Bottem up parsing can be determined similar to that of TOP Down parsing with the change that in this case ,the process is started with the given string and the tree is extended till "S "is Regular

- Ø Non regular
- Ø **Obtain** **Page 146**
- Ø Finite

NO127: A production in CFG consists of:

- Ø One terminal
- Ø More than one terminal
- Ø One non-terminal
- Ø **Terminals and non-terminals**

NO128: If L1 and L2 are regular languages is/are also regular language(s).

- o L1 + L2
- o L1L2
- o L1*
- o **All of above**

NO129: Between the two consecutive joints on a path

- Ø One character can be pushed and one character can be popped

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- Ø **Any no. of characters can be pushed and one character can be popped**
- Ø One character can be pushed and any no. of characters can be popped
- Ø Any no. of characters can be pushed and any no. of characters can be popped

NO131: Language which are context –free are called Non –CFL:

- Ø True
- Ø **False** **Page 132**

NO132: The following problem(s) ---------- is/are called decidable problem(s).

The two regular expressions define the:

- Ø same language
- Ø The two FAs are equivalent
- Ø **Both a and b**
- Ø None of given

NO133: If F accept an ……. language then there are some words w.s.t N≤ length (w) <2n :

- Ø Regular
- Ø Finite
- Ø **Infinite** **Page 91**
- Ø None of given

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NO134:If an FA has N state then it must accept the word of length:

- Ø N^2
- Ø N-1
- Ø **n+1**
- Ø all of above

NO135: The values of input (say a & b) does not remain same in one cycle due to

- Ø tape
- Ø halt
- Ø **clock pulse**
- Ø start

NO136: Consider the following CFG
s→aS|bS|aaS|Λ
can be observed that the word aaa can be derived from more than:

- Ø **one production tree** **Page 101**
- Ø two producton tree
- Ø Total language tree
- Ø All of above

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NO137: One language can have …… CFG(s)

- Ø **At least one** Ø
- At least two Ø At least three
- Ø None of them

NO138: The reverse of the string sbfsbb over { sb, f, b}

- Ø **(bsbfsb)**
- Ø bfsbs
- Ø sbbfsb
- Ø bbfsb

NO139: CFG is said to be a regular grammar if it generates the regular language i.e.a CFG is said to be a regular grammar in which each production is one of the:

- Ø Three forms
- Ø One form
- Ø Four forms
- Ø **Two forms Page 102**

NO140: If L1 and L2 are regular languages then which statement is NOT true?

- Ø **(L1/L2 is always regular)** Ø
- L1+L2 are always regular Ø
- L1*l2 are always regular
- Ø None of them

NO141: If the intersection of two regular languages is regular then the complement of the intersection of these two languages is also regular:

- Ø False
- Ø **True**

NO142: the moment a final state of ……. FA is entered ,the possibility of the initial state of ……. FA will be included as well:

- Ø first ,third
- Ø **first ,second Page 39**
- Ø second ,third
- Ø all of above
- Ø

NO143: Any word generated by given CFG cannot also be expressed by Syntax tree or Generation tree or Derivation tree as well

- Ø true
- Ø **false**

NO144: According to Mayhill Nerode theorem, if L generates finite no. of classes then L is…..

- Ø **Regular**

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- Ø Nonregular
- Ø Infinite
- Ø Finite

NO145: L is a regular language so by kleene,s theorem ,there exists an:

- Ø **FA Page 71**

- Ø GTG
- Ø TG
- Ø CNF

NO146: The language generated by the CFG is called the language ……by the CFG

- Ø **Produced**
- Ø Null string
- Ø Pumping lemma
- Ø Non of then

NO147: In CFG, the symbols that cannot be replaced by anything are called:

- Ø None terminals
- Ø Infinite
- Ø Finite
- Ø **Terminals**

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NO148: The production S --> SS | a | b | ^ can be expressed by RE:

- Ø **(a+b)+**
- Ø a-b
- Ø (a-b)+
- Ø None of them

NO149: Set of all palindromes over {a,b}is regular

- Ø **(false)**
- Ø (true)

NO150: An FA has same initial and final state, then it means that it has no final state.

- Ø **(false)**
- Ø (true)

NO151: The same non terminals can be written in single line if they have more than one.....

- Ø **(Productions)**
- Ø Regular production
- Ø None regular production

NO152: If L1 and L2 are two regular languages then L1∩L2 is also :

- Ø **Regular Page 73**

- Ø None regular
- Ø Finite
- Ø None infinite

NO154: For language L defined over {a, b},then L partitions {a, b}* into …… classes:

- Ø **(Distinct)**
- Ø Accept
- Ø Unit production
- Ø None of the above

NO155: The two FAs are

- Ø Same
- Ø **Equivalent** **Page 85**
- Ø Different
- Ø None of them

NO156:There is at least one production that has one……..on its left side:

- Ø Terminal
- Ø Infinite
- Ø **None Terminal**
- Ø All of above

NO157:The complement of a regular language is also a regular

- Ø **(True)**
- Ø False

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- Ø **NO159: If an effectively solvable problem has answer in yes or no ,then this solution is called : Decision procedure** **Page 85**
- Ø Decidable problem
- Ø Solved able problem
- Ø All of above

NO160: In pref(Q in R) Q is …… to (than) R

- Ø **Q is not equal to R**
- Ø Q is equal to R
- Ø Q is infinite
- Ø None of them

NO61: For FA corresponding
To $(L1 \cap L2^{C}) U (L1^{C} \cap L2)$ the regular expression ca be determined that defines the language accepted by this :

- Ø TG
- Ø GFC

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- Ø **FA** **Page 85**
- Ø GTG

NO162: a^n b^n generates the …… language:

- Ø **Non regular languages**
- Ø **Regular language** Ø
- **Infinite language** Ø
- **Finite language**

NO: 163 (a+b)*a(a+b)*b(a+b)* is the RE of language defined over ={a,b} having at least one a and one b

- Ø **True** **Page 11**
- Ø False

Such a language does not exist
None of these

Question No: 164: (Marks: 1) - Please choose one
NO164: If r1 = (aa + bb) and r2 = (a + b) then the language (a + b)* (aa + bb)* will be generated by :

- Ø (r2)(r1)
- Ø (r1 + r2)*
- Ø (r2)*(r1)*
- Ø **(r1)*** **Page 12**

Question No 165: (Marks: 1) - Please choose one
NO165: In FA starting state is represented by a _____sign.

- Ø **+**
- Ø **-**
- Ø *****
- Ø **S**

NO166: If w is large enough word in a CF then w can be decomposed into w =uvwxyz such that all words of the form uvn xynz belong to :

- Ø CNF
- Ø **L**
- Ø CFL
- Ø CFG

NO167: Can a turing machine's head ever be in the same location in two successive steps?

- Ø Yes
- Ø Yes but only in finite languages
- Ø **No** **Page 150**
- Ø Yes but only in infinite langauges

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NO168: Examin the following CFG and select the correct choice:
S→ AB, A →BSB, B→ CC
C
→

S
S
A
→

a
|
b
C
→

b
|
b
b

- Ø **abb is a word in the corresponding CFL.** **Page 139**
- Ø abb is not the word of corresponding CFL.
- Ø any word can be accept from the corresponding CFL.
- Ø Non of these

NO169: The production of the form nonterminal →string of two nonterminals is called a:

- Ø **live production** **Page 132**

- Ø dead production
- Ø type of production
- Ø none of them

Question No: 10 (Marks: 1) - Please choose one
NO170: Converting the given CFG in CNF is the first rule of _____

- Ø **CYK algorithm Page 140**
- Ø CKY algorithm
- Ø KYC algorithm
- Ø CNK algorithm

Question No: 11 (Marks: 1) - Please choose one
NO171: Which statement is true?

Ø The PDA must have one accept state and one reject state Ø The PDA must have one accept state and two reject state Ø The PDA must have two accept state and two reject state Ø **There is no reject state in the PDA.Page 124**

NO172: Question No: 12 (Marks: 1) - Please choose one

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If a language can be expressed by a regular expression, then its complement cannot be expressed by a regular expression. This statement is:

- Ø true
- Ø **False**
- Ø Depends on language
- Ø None of the given optios

NO173: Left hand side of CFG may consist of:

- Ø One terminal
- Ø More than one terminal
- Ø **One non-terminal**
- Ø Terminals and non-terminals

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